

State or Signal? Observable-Relative Causal Aliasing under Receiver-State-Dependent Relabeling

Ruben Brito

Independent researcher
ORCID: 0009-0003-3276-9113

Abstract

Interventional invariance is often treated as evidence that a causal organization is preserved across contexts. Whether that inference is warranted, however, depends on which aspects of causal response survive the chosen representation. We preregistered a confirmatory test of causal individuation in a controlled finite-state system designed to distinguish direct interface use from receiver-state-dependent interface remapping, with confirmatory execution conducted under blinding. The confirmatory result was negative: the primary response-order criterion matched all 150 base-stratum targets but 49 of 50 downstream-extension targets, thereby failing the preregistered conjunctive criterion. Post-confirmatory analysis of the sole counterexample established receiver-memory-dependent causal response differences on common realized support, ruling out a purely architectural explanation for the collapse. We then identified the mechanism. Receiver state induces contextual relabelings of the interface, and the relative relabeling acts on a mechanism-induced response geometry. If the retained observable is invariant under that transformation, causally distinct receiver contexts can become observationally equivalent at the level used for individuation. For the frozen intervention design, exact-profile blindness admits an exact symmetry characterization; a separate ordinal blind route occurs when cardinal response profiles differ while their weak order is preserved. Across 20,000 decisive edges generated under the frozen eligible distribution, progressively finer response representations strictly reduced the observed indistinguishability set; complete two-edge collapse occurred in 46 of 10,000 eligible System-II instances. The result is not a claim of universal causal non-identifiability or a universal state-signal ontology. It establishes a concrete failure mode of interventional causal individuation: receiver-state-dependent relabeling can preserve the observable while altering the underlying causal response field.

Keywords: causal inference; causal individuation; causal abstraction; observability; symmetry; identifiability; finite-state systems; interventional invariance

Epistemic status of the evidence

Layer	Status in this manuscript
preregistered confirmatory test	negative ; frozen scoring unchanged
family-20 mechanism and realized-support analysis	post-confirmatory, demonstrated and internally reproduced
exact-profile theorem and alignment theorem	post-confirmatory analytic results with finite-domain verification
generator-specific frequency and refinement counts	post-confirmatory empirical/census results under the frozen generator
biology/AI applications and system-observable-horizon synthesis	prospective or programmatic only

Scope note. STATE and SIGNAL are prespecified operational labels of the synthetic testbed, not universal natural kinds. The term *observable-relative causal aliasing* is defined in this paper for $X_0 \neq X_1$ together with $F(X_0) = F(X_1)$; observable equality alone is only blindness.

Contents

Abstract	1
1. Introduction	4
2. Results	5
2.1 The preregistered evaluation is negative	5
2.2 The response-order counterexample contains a causal difference on common realized support	6
2.3 Receiver state induces a contextual pull-back of the response field	7
2.4 Mechanism-dependent symmetry creates exact-profile blindness	8
2.5 The frozen intervention family makes the exact-profile blind route exact	10
2.6 The response-order criterion also admits an ordinal blind route	11
2.7 Observable-induced indistinguishability shrinks under refinement	12
2.8 Generator-specific frequency under the frozen generator	13
3. Discussion	14
3.1 What failed?	14
3.2 What was characterized?	14
3.3 What else does the response-order criterion lose?	15
3.4 What is not being claimed?	15
3.5 Why does it matter?	16
4. Methods	18
4.1 Operational targets and construct validity	18
4.2 Preregistered protocol, generation, and blinding	19
4.3 Interventions, one-step interventional influence graph, response-order criterion, and confirmatory scoring	20
4.4 Response-field definitions for the post-confirmatory analysis	23
4.5 Response-sufficient local-state comparability and realized differences	23
4.6 Contextual pull-back lemma for System II	24
4.7 Mechanism-induced response pseudometric	25
4.8 Class-level symmetry-blindness proposition	25
4.9 Exact characterization of the exact-profile blind route	26
4.10 Ordinal blindness and the alignment theorem	28
4.11 Strict-realization analysis and witness extraction	29
4.12 Information-loss refinement hierarchy	31
4.13 Post-confirmatory generator-specific frequency study	32
4.14 Temporal-composition analysis	34
4.15 Internal verification, provenance, and reproducibility	37
5. Construct validity and scope	40
The target distinction is operational and fixed by construction	40
The counterexample is not only an architectural label mismatch	40
The result concerns an observable, not all causal information	40
The analytic result has an explicit theorem class	41
Internal validity is stronger than external validity	41
Generalization and falsification require separate studies	42
Implication for interpretation	42
6. Relation to prior work	42
Symmetry-induced unobservability is established prior art	42
Lossy causal abstraction and aggregation are also established	43
Endogenous contexts can alter causal mechanisms	43
Micro-realization dependence and the realized witness	44
Finite-horizon distinguishability is not new	44
Restricted injectivity and orbit separation are established mathematical structures	44
The alignment theorem is instrument-specific	44

Specific gap retained after the adversarial audit	45
Positioning statement	45
7. Limitations	46
The confirmatory test was negative, whereas the central mechanism was characterized post-confirmatorily	46
The testbed is synthetic and its operational targets are fixed by construction	46
The analytic theorem has a restricted structural domain	46
The strict-realization result is local to the analyzed contrast	47
The empirical frequencies are conditional on one eligible generator law	47
Only C1' received a mechanistic failure analysis	47
The blinding was procedural rather than cryptographic	47
Internal reproduction is not external replication	48
Observable refinement is not guaranteed to solve identification	48
Temporal conclusions are narrow	48
Novelty remains a candidate, not a proved historical priority	48
Real-world relevance remains prospective	49
Author contributions and data availability	49
References	49
Supplementary Note A. Formal status and notation	51
Supplementary Note B. Local-response reduction and realized support	51
Supplementary Note C. Proof of Proposition 1 (symmetry-induced blindness)	52
Supplementary Note D. Measurement-operator criterion, exact-profile theorem, and frozen-design rank audit	52
Supplementary Note E. Ordinal blindness and the alignment theorem	53
Supplementary Note F. Information-loss hierarchy	54
Supplementary Note G. Generator-specific frequency study	54
Supplementary Note H. Temporal composition in the direct-interface benchmark	54
Supplementary Note I. Frozen source manifest and execution provenance	55
Supplementary Note J. Evidence-status ledger for the main claims	55

1. Introduction

Interventions can establish that changing one part of a system changes another, but establishing influence does not by itself settle every mechanistic distinction an analyst may wish to draw from that evidence. In systems with internal state, the same sender-receiver pair can be embedded in different receiver contexts, and an operational criterion must decide whether those contexts preserve the causal relation **according to the representation and target being tested** or instantiate a change that the criterion is intended to distinguish. A natural strategy is to compare interventional response profiles across contexts and treat preserved response structure as evidence of causal invariance. The central methodological question is therefore not only whether an intervention changes an outcome, but whether the **representation used to summarize that intervention retains the distinctions needed for the intended mechanistic classification**.

There are strong reasons not to assume that such representations are automatically identifying. Symmetry can make distinct states or trajectories observationally indistinguishable (Whalen et al., 2015; Kolar et al., 2018; Mesbahi et al., 2021). Causal abstraction and aggregation can merge low-level distinctions or make macro-level causal interpretation depend on micro-realization (Zhu et al., 2024; Xia & Bareinboim, 2025). Internal or endogenous context can also change causal mechanisms or context-specific causal relations (Günther et al., 2024), while interventional data may identify latent causal structure only up to an equivalence or abstraction determined by the available interventions (von Kügelgen et al., 2023; Li et al., 2025). These literatures establish the generic possibility of information loss and symmetry-induced unobservability. They do not by themselves determine when receiver internal state can generate a context-dependent causal change that is subsequently erased by the particular interventional observable used to detect it.

The problem was addressed in a controlled synthetic setting. The preregistered design fixed operational interaction targets rather than assuming universal natural kinds. In the critical contrast, the **direct-interface benchmark (System III)** presents an incoming interface directly to the receiver mechanism, whereas the **receiver-state-remapped interface system (System II)** allows retained receiver state to remap that interface before the receiver mechanism is evaluated. System I is a **direct-coupling benchmark** without the explicit interface modules. Candidate causal criteria intended to preserve the corresponding component partitions under frozen intervention, scale, and representational-equivalence rules were preregistered and blindly evaluated. Full confirmatory outcomes for C1', C2, and C3 are reported in Results. The mechanistic analysis follows the **interventional response-order invariance criterion (C1')** because its single confirmatory counterexample revealed the mechanism studied here. C1' classifies a receiver-facing edge by whether its interventional response profile preserves the same canonical weak order across receiver-memory contexts. C2 is the **effective-intervention partial-order invariance criterion**; C3 is a **support-invariance implementation canary** and does not enter the scientific hypothesis decision.

The confirmatory test was **negative**. C1' matched the preregistered targets in 150/150 E1 instances and 49/50 E2 instances, but the E2 target was conjunctive and therefore failed; the official result remained **passa=false**. The aggregate 199/200 count is not interpreted as validation or near success. Instead, the single C1' confirmatory counterexample posed the question that motivates the present article: **why did an interventional invariance criterion fail on a system constructed under the preregistered System-II target?**

A trivial answer would have been that the failed target encoded only an architectural distinction and that the realized dynamics contained no causal difference for the criterion to recover. Post-confirmatory analysis ruled out that explanation for the C1' counterexample. On both decisive receiver-facing edges, receiver memory changed the underlying response field; both memory contexts were reached; and the difference was present in realized dynamics at the response-sufficient local state (r, c) for the frozen one-step contrast. Thus the relevant situation is not merely

$$F(X_0) = F(X_1),$$

which establishes blindness of an observable, but

$$X_0 \neq X_1 \quad \text{and} \quad F(X_0) = F(X_1),$$

with the difference separately demonstrated on realized support. This paper uses **observable-relative causal aliasing** as an explicitly defined descriptive term for this stronger situation; it is not assumed to be standard causal-inference nomenclature.

The post-confirmatory mechanism can be stated compactly. Receiver memory induces contextual interface relabelings π_m , with relative interface permutation

$$\tau = \pi_1 \circ \pi_0^{-1}.$$

The receiver mechanism induces a pairwise response pseudometric W_M . For the relevant class of cell-factorizing observables, a relative interface permutation that lies in the isometry group of that pseudometric is invisible. In the frozen nine-intervention family, the exact-profile blind route is characterized exactly by

$$d_0 = d_1 \iff \widetilde{W}_0 = \widetilde{W}_1 \iff \tau \in \text{Iso}(W_M).$$

The generic mathematics of symmetry invariance is established prior art; the causal content here comes from the composition in which **receiver internal state generates the transformation, the concrete receiver mechanism determines the geometry with respect to which it is symmetric, and the resulting measured invariance coexists with a separately demonstrated realized causal difference**. Our adversarial prior-art audit did not identify an essentially equivalent antecedent for that complete composition, but this is a novelty candidate rather than a claim of proved historical priority.

C1' also exposes a second blind route. Its final decision uses weak-order rather than cardinal equality, so two contexts can have

$$d_0 \neq d_1$$

while retaining the same weak rank. This ordinal L2 route is distinct from the exact-profile isometry route. More generally, the post-confirmatory refinement analysis shows that the blind set depends on what response information the observable retains. Point-pattern, row-resolved, multiset-resolved, and cell-aggregated representations are ordered by refinement, and the corresponding blind sets were observed to be strictly nested in the studied population. The result therefore does not establish global causal non-identifiability from all available causal information. It establishes that a causal difference can be non-identifiable **relative to a particular interventional representation even when finer causal information remains separating**.

This article makes four headline contributions. First, it reports the preregistered negative result without reinterpretation. Second, it gives an exact characterization of a symmetry-induced exact-profile blind route for the frozen nine-intervention family and a broader one-way blindness proposition for the stated contextual-pullback class. Third, it demonstrates that the confirmatory counterexample contains a causal difference that is present on common realized support at a response-sufficient local state, distinguishing causal aliasing from observable equality alone. Fourth, it characterizes how ordinal coarsening and observable refinement alter the blind set. Together these results turn a failed discriminator into a bounded analysis of **when, in the studied class, measured interventional invariance can fail to track an underlying causal difference**.

2. Results

2.1 The preregistered evaluation is negative

A blinded confirmatory evaluation of causal-individuation candidates on synthetic systems assigned by construction to prespecified operational interaction classes was preregistered. The final frozen v8.3 executable instrument scored C1', C2, and C3. C1' and C2 were the candidates capable of making the scientific hypothesis H_A positive; C3 was an implementation canary and did not enter H_A . No C4 path was executable or scored in the frozen confirmatory instrument.

The confirmatory outcome was negative. For C1', all 150 E1 instances matched their preregistered targets, whereas 49 of 50 E2 instances did so. Thus C1' yielded 199/200 target matches overall, but failed the preregistered conjunctive success requirement because E2 was not perfect. The official result therefore remained **passa = false**; the numerical proportion is not interpreted as a near-positive result. C2 disagreed with its preregistered target in 74/200 instances, and C3 in 75/200. Both also disagreed on the instance that produced the single C1' error. C2 and C3 are reported for completeness but were not subjected to the mechanistic post-confirmatory autopsy developed below.

The single C1' confirmatory counterexample was instance 7bb0baab3a8ed7aa, family 20, a System-II instance in E2/Stratum 2. C1' fused the core $\{A, B, C_{AB}, C_{BA}\}$ into one strongly connected STATE component, producing the operational partition associated with System III rather than the preregistered System-II target. This classification was a genuine confirmatory error under the frozen target and was not altered after scoring.

All analyses from this point onward are post-confirmatory. They were performed after the official result had been closed, without changing the criterion, targets, thresholds, scoring, or confirmatory artifacts. No replacement criterion C1'' was formulated. The question addressed by the autopsy was therefore explanatory rather than corrective: **why did C1' fail on a system constructed under the preregistered System-II target?**

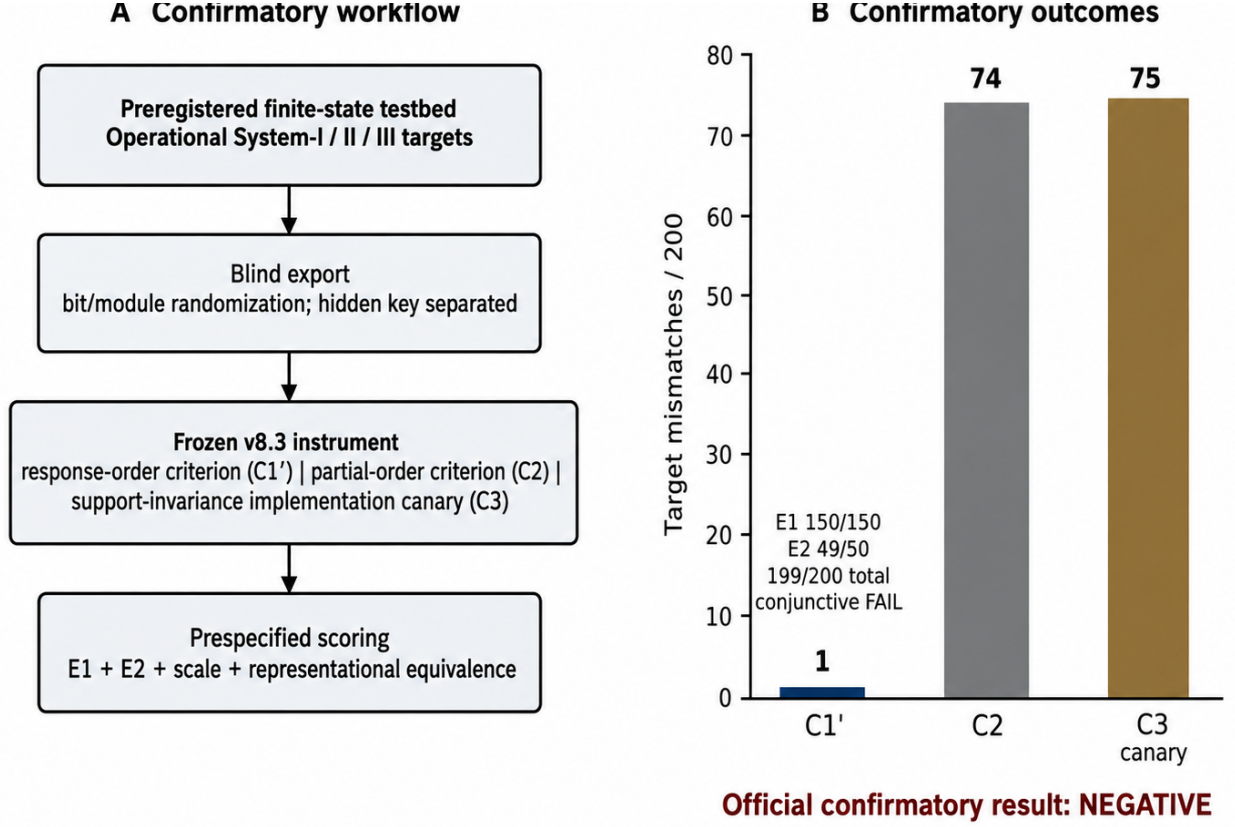


Figure 1. Preregistered workflow and confirmatory outcome. The scientific result is negative because the E2 requirement was conjunctive. C3 is a support-invariance implementation canary and does not enter H_A .

2.2 The response-order counterexample contains a causal difference on common realized support

A first possibility was that the System-II construction differed from System III only architecturally, while the causal distinction required by the target was absent from the realized dynamics of family 20. The post-confirmatory analysis rejected this explanation.

For each of the two decisive channel-to-processor edges, $C_{AB} \rightarrow B$ and $C_{BA} \rightarrow A$, the full interventional response field contained 4,608 sites whose response depended on receiver memory. Both memory contexts were reached in the realized orbit: the orbit contained 15 versus 10 states for the two B -memory contexts on $C_{AB} \rightarrow B$, and 12 versus 13 states for the two A -memory contexts on $C_{BA} \rightarrow A$. Thus the memory contexts responsible for the comparison were not inaccessible counterfactual states.

To ask whether the difference occurred in response-comparable states actually visited under both contexts, the analysis used the local state (r, c) , where r is the receiver's non-memory processor state and c is the channel value. Conditional on receiver memory, the one-step interventional response is a function only of (r, c) . Consequently, (r, c) is a response-sufficient local state for this contrast. Refining it with globally irrelevant bits can remove overlap between contexts without changing the response being compared; coarsening can merge locally distinct responses, so no stronger minimality claim is needed.

At this granularity, both decisive edges contained strictly realized memory-dependent differences. For $C_{AB} \rightarrow B$, four (r, c) cells were visited under both receiver-memory contexts and nine intervention-by-cell sites differed across those contexts. For $C_{BA} \rightarrow A$, five shared cells likewise contained nine differing sites. The result did not rely on

full global configurations being identical across contexts.

The distinction was also visible using realized transitions alone. On each decisive edge, pairs of realized transitions with matched local receiver state yielded different observational contrasts across memory contexts. Therefore, the C1' collapse cannot be explained by an absence of realized separating causal information. A causal difference existed below the aggregate that C1' used.

This establishes the empirical premise required for the interpretation below. For any observable F , equality $F(X_0) = F(X_1)$ defines observable-induced indistinguishability. The shorthand **blindness** is used for that equality. The stronger term **observable-relative causal aliasing** is reserved for cases in which an underlying causal difference $X_0 \neq X_1$ has independently been established. Family 20 satisfies this stronger condition, including at the response-sufficient local state.

A Confirmatory counterexample architecture

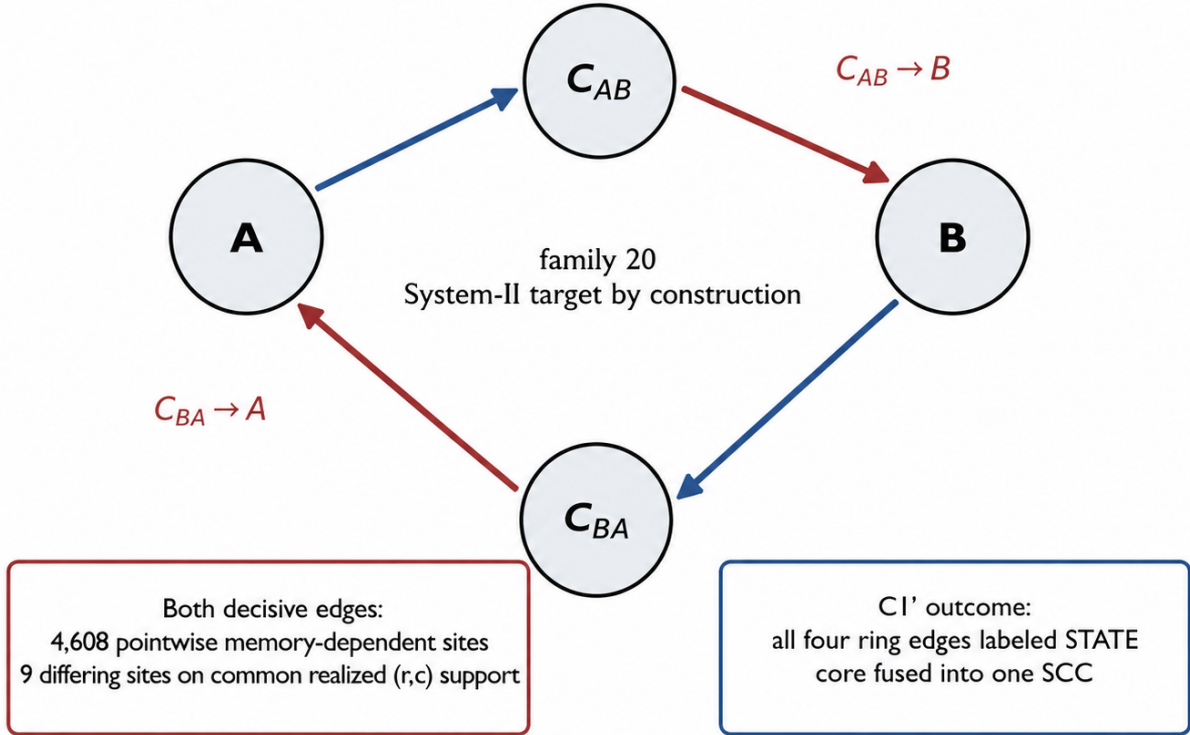


Figure 2. Family 20, the single response-order confirmatory counterexample. Both decisive channel-to-processor edges show receiver-memory-dependent response differences, including nine differing intervention-by-cell sites on common realized (r, c) support per edge, yet all four ring edges are labeled STATE by the frozen response-order criterion.

2.3 Receiver state induces a contextual pull-back of the response field

The failure can be reduced analytically to how receiver memory enters the frozen one-step channel-to-processor contrast under the aligned full-fiber comparison. For the relevant System-II dynamics, the receiver response can be written in the form

$$M[r][\pi_m(c)] \oplus \sigma[m],$$

where $m \in \{0, 1\}$ is receiver memory, r is the receiver's non-memory processor state, c is the channel symbol, π_m is a memory-dependent relabeling of the interface, and $\sigma[m]$ is an additive context term.

For an intervention a that replaces c by $\text{sub}_a(c)$, the XOR response contrast is therefore

$$X_m(a; r, c) = M[r][\pi_m(c)] \oplus M[r][\pi_m(\text{sub}_a(c))],$$

with $M = G_0$ for $C_{AB} \rightarrow B$ and $M = F_0$ for $C_{BA} \rightarrow A$. The additive term $\sigma[m]$ cancels from the XOR contrast.

The receiver-memory update is independent of the intervened channel at the current step and consequently does not contribute to the one-step response difference. Memory nevertheless remains causally relevant because it changes the interface relabeling π_m .

For this one-step contrast, the two memory contexts are therefore pull-backs of a common response mechanism under different relabelings of the same finite interface. Their relative interface permutation is

$$\tau = \pi_1 \circ \pi_0^{-1}.$$

This is the key structural reduction. The relevant question is no longer simply whether memory changes the pointwise response—it does in family 20—but whether the observable used by C1' is sensitive to the particular relabeling induced by that memory change.

2.4 Mechanism-dependent symmetry creates exact-profile blindness

Define the pairwise response pseudometric induced by receiver mechanism M as

$$W_M(p, q) = \sum_r \text{pc}_2(M[r][p] \oplus M[r][q]),$$

where pc_2 is the Hamming popcount on the two-bit response. Each row contributes the pull-back of a Hamming metric and the sum is therefore a pseudometric: distinct interface symbols may have zero distance if the receiver mechanism never distinguishes them. Receiver memory acts on this pseudometric by relabeling its arguments through π_m .

Proposition 1 (Symmetry-induced blindness for contextual pull-backs). *Let two aligned contextual response fields be pull-backs of the same finite response tensor under interface relabelings π_0 and π_1 . Let $\widetilde{W}_m(c_1, c_2) = W_M(\pi_m(c_1), \pi_m(c_2))$, and suppose an observable F factors through $\widetilde{W}_m$. If the relative interface permutation $\tau = \pi_1 \circ \pi_0^{-1}$ belongs to $\text{Iso}(W_M)$, then $F(\widetilde{W}_0) = F(\widetilde{W}_1)$. This proposition establishes blindness only; causal aliasing additionally requires an independently established difference in the underlying response fields.*

This yields a class-level blindness proposition under the comparison conditions used here. Suppose two aligned contextual response fields are pull-backs of the same finite response tensor under relabelings π_0 and π_1 , and suppose the observable factors through the cell-weight geometry induced by that response tensor. If the relative relabeling

$$\tau = \pi_1 \circ \pi_0^{-1}$$

is an isometry of that geometry,

$$\tau \in \text{Iso}(W_M),$$

then the observable is invariant across the two contexts. This is a symmetry statement about the representation measured by the observable; by itself it does not establish that the underlying fields differ.

A minimal post-confirmatory toy illustrates the logical distinction between blindness and causal equality without relying on the confirmatory generator. Let $C = \{0, 1, 2\}$, with two receiver rows and one-bit outputs,

$$M[0] = (0, 0, 1), \quad M[1] = (0, 1, 0),$$

and let $\pi_0 = \text{id}$, $\pi_1 = (12)$. Probe the three full-channel interventions $\text{do}(c := v)$, $v \in \{0, 1, 2\}$, and let the resulting aggregate profile be the corresponding three degree sums $V(v) = \sum_{u \in qv} W(v, u)$. The induced geometry has

$$W(0, 1) = 1, \quad W(0, 2) = 1, \quad W(1, 2) = 2,$$

so $\text{Iso}(W) = \{\text{id}, (12)\}$ and the relative interface permutation $\tau = (12)$ is the non-trivial isometry. Nevertheless, the pointwise response fields differ—for example, one matched intervention/site changes from response pattern $(0, 1)$ to $(1, 0)$ —while the aggregate profiles are identical:

$$d_0 = d_1 = (2, 3, 3).$$

The toy is illustrative only; it is not confirmatory evidence and does not establish realized dynamics for family 20.

Family 20 supplies the confirmatory counterexample in which the same logical structure coexists with an independently established, causal difference on common realized support. On $C_{BA} \rightarrow A$, all off-diagonal distances in W_M equal 4, so

$$\text{Iso}(W_M) = S_4.$$

Every permutation of the four interface symbols is therefore invisible at this geometric level. On $C_{AB} \rightarrow B$,

$$W_M = \begin{pmatrix} 0 & 4 & 5 & 5 \\ 4 & 0 & 3 & 3 \\ 5 & 3 & 0 & 4 \\ 5 & 3 & 4 & 0 \end{pmatrix},$$

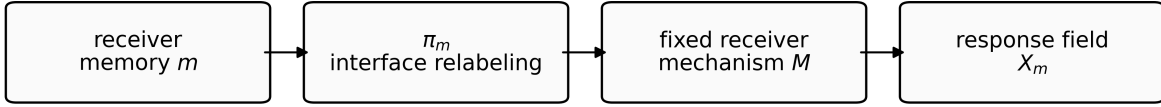
for which

$$\text{Iso}(W_M) = \{\text{id}, (23)\}.$$

The relative interface permutation realized by family 20 is exactly the non-trivial transposition (23). Thus the same relative interface permutation lies in the isometry group of both decisive receiver geometries: trivially on the equidistant edge and selectively on the second edge. Yet Section 2.2 established that the underlying pointwise response fields differ and that those differences are strictly realized.

Family 20 therefore realizes **observable-relative causal aliasing**: a receiver-state-dependent causal modulation is present, but the relabeling through which it acts lies inside a symmetry of the mechanism-induced response pseudometric retained by the aggregate.

A Contextual pull-back and measurement



$$\text{relative interface permutation: } \tau = \pi_1 \circ \pi_0^{-1}$$

$$W_M(p, q) = \sum_r \text{pc}_2(M[r][p] \oplus M[r][q])$$

B Family-20 response pseudometrics

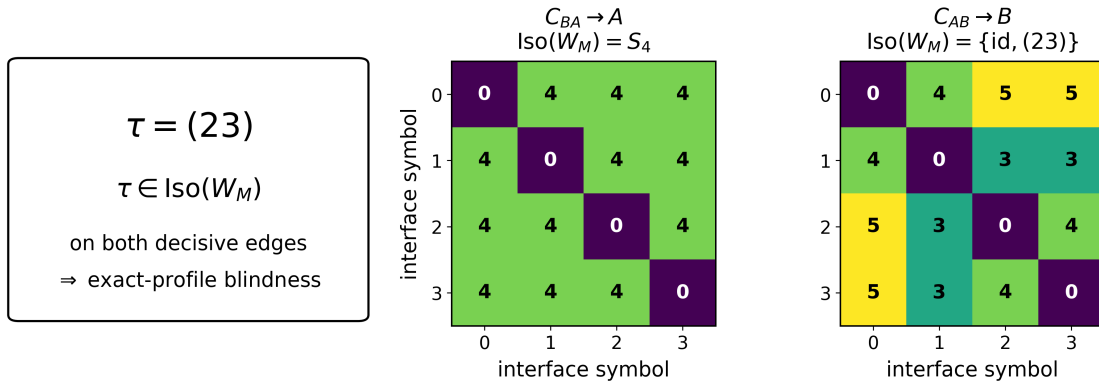


Figure 3. Receiver-state relabeling and exact-profile blindness. The relative interface permutation is $\tau = \pi_1 \circ \pi_0^{-1}$. In family 20, $\tau = (23)$ lies in the isometry group of the mechanism-induced response pseudometric on both decisive edges.

2.5 The frozen intervention family makes the exact-profile blind route exact

The symmetry condition is not only sufficient for exact-profile blindness in the frozen instrument. Its dependence on the intervention design can be separated from the receiver mechanism by treating the interventional summary as a measurement operator.

The following operator-level statement is a post-confirmatory formal extension and uses no new empirical data. Let $\mathcal{V}_C$ denote the vector space of symmetric, zero-diagonal pairwise arrays on a finite interface alphabet C , and let $\mathcal{A}_{\mathcal{I}} : \mathcal{V}_C \rightarrow \mathbb{R}^k$ be the linear measurement operator induced by an intervention family $\mathcal{I}$. For a mechanism-induced pseudometric W , define the relabeling action by

$$(\pi^*W)(c_1, c_2) = W(\pi(c_1), \pi(c_2)),$$

and its relabeling orbit by

$$\mathcal{O}(W) = \{\pi^*W : \pi \in S_C\}.$$

and its orbit-difference set

$$\Delta\mathcal{O}(W) = \{U - V : U, V \in \mathcal{O}(W)\}.$$

Proposition 2 (Measurement-operator criterion for exact-profile identification). *If the contextual summaries satisfy $d_m = \mathcal{A}_{\mathcal{I}}(\widetilde{W}_m)$, then*

$$d_0 = d_1 \iff \widetilde{W}_1 - \widetilde{W}_0 \in \ker(\mathcal{A}_{\mathcal{I}}).$$

Moreover, the restriction of $\mathcal{A}_{\mathcal{I}}$ to the relabeling orbit $\mathcal{O}(W_M)$ is injective if and only if

$$\ker(\mathcal{A}_{\mathcal{I}}) \cap \Delta\mathcal{O}(W_M) = \{0\}.$$

Under this equivalent orbit-separation condition, exact-profile equality implies equality of the contextual pseudometrics, and hence

$$d_0 = d_1 \iff \widetilde{W}_0 = \widetilde{W}_1 \iff \tau \in \text{Iso}(W_M).$$

Thus the exact converse does not depend on the number nine as such; it depends on whether the chosen intervention operator separates the relabeling orbit relevant to the mechanism.

Corollary 1 (Ambient dimensional requirement). *For an interface alphabet of size q , the ambient space of symmetric, zero-diagonal pairwise arrays has dimension $\binom{q}{2}$. A linear measurement operator that is injective over that full ambient space must therefore have rank $\binom{q}{2}$ and at least that many independent scalar measurements. For the frozen four-symbol interface, $\binom{4}{2} = 6$.*

The frozen nine-intervention family induces an injective linear measurement map for the six independent entries of the induced pairwise pseudometric and therefore satisfies this stronger ambient-space condition.

Let

$$\widetilde{W}_m(c_1, c_2) = W_M(\pi_m(c_1), \pi_m(c_2)).$$

For the frozen statistic,

$$d_m(a) = 2^{n-5} \sum_c \widetilde{W}_m(c, \text{sub}_a(c)).$$

Equivalently, after factoring out the common multiplicity, the nine-component profile has the structural form

$$d = 2^{n-5}(0, A, A, B, B, V),$$

where $V = (V_0, V_1, V_2, V_3)$ denotes the four row-sum components; together with the two bit-intervention pairing sums, these provide six independent measured degrees of freedom. The map from the six off-diagonal coordinates of $\widetilde{W}_m$ to the nine-component profile d_m is injective (an isomorphism onto its image). After removing the null and duplicated coordinates, the reduced 6×6 measurement matrix L is invertible, with rank 6 and determinant -8 ; the two restricted intervention subfamilies examined have ranks 2 and 4 and do not support the converse.

A post-confirmatory exact rank audit of all $2^9 = 512$ subsets of the frozen intervention family further shows that the full-rank property is not contingent on retaining all nine interventions. Exactly 18 subsets have rank 6. Four are minimal six-intervention subsets: each contains all four full-channel assignments $c := 0, 1, 2, 3$, one of the two duplicated low-bit assignments, and one of the two duplicated high-bit assignments. This is a property of the frozen measurement design, not a new empirical result.

Theorem 1 (Exact-profile blindness for the frozen intervention family). *For a frozen channel-to-processor edge under the complete nine-intervention family,*

$$d_0 = d_1 \iff \widetilde{W}_0 = \widetilde{W}_1 \iff \tau \in \text{Iso}(W_M).$$

The equivalence characterizes the exact-profile blind route (L1) only. The full STATE event of the response-order criterion also includes ordinal blindness (L2).

Consequently, for the frozen nine-intervention family,

$$d_0 = d_1 \iff \widetilde{W}_0 = \widetilde{W}_1 \iff \tau \in \text{Iso}(W_M)$$

for the relevant channel-to-processor edges.

This equality event is denoted **L1**, the exact-profile blind route. The result is exact for this instrument: once the response has been reduced to $\widetilde{W}$, the map from $\widetilde{W}$ to d introduces no additional L1 information loss. The theorem does **not** characterize every STATE decision made by C1', because C1' subsequently replaces cardinal profiles by their weak order. That second route is addressed next.

2.6 The response-order criterion also admits an ordinal blind route

C1' does not compare d_0 and d_1 directly. It assigns the relevant edge to STATE when their canonical weak ranks coincide. This yields two distinct cases:

$$\text{L1} : d_0 = d_1,$$

and

$$\text{L2} : d_0 \neq d_1 \quad \text{but} \quad \text{rank}(d_0) = \text{rank}(d_1).$$

Both produce the same STATE edge label under C1'. An instance-level System-II collapse requires both decisive channel-to-processor edges to receive that label, with either edge allowed to reach it through L1 or L2.

L2 is not a weakened version of the exact isometry condition. In the primary 10,000-family population, 225 of 20,000 decisive edges were L2. Among the 24 L2 edges contained in the 46 total-collapse instances, 17 had a trivial exact isometry group $\text{Iso}(W_M)$. Their cardinal profiles changed; what remained invariant was the weak-order representation consumed by C1'.

Theorem 2 (Alignment constraint on ordinal blindness). *Let $\psi : S_4 \rightarrow S_3$ be the induced action of interface relabelings on the three perfect matchings of the four-symbol interface, and let λ denote the matching not directly probed by the bit interventions. If $\psi(\tau)$ fixes λ , equality of the canonical weak-rank profiles implies $d_0 = d_1$. Hence L2 is impossible in the aligned region; ordinal blindness requires a misalignment that moves the unprobed matching into a measured position while preserving weak order.*

The ordinal blind zone can nevertheless be characterized relative to the frozen intervention geometry. The four interface symbols have three perfect matchings, while the bit interventions directly probe only two. Let

$$\psi : S_4 \rightarrow S_3$$

be the action of interface relabelings on those three matchings, with kernel V_4 , and let λ denote the unprobed matching. If $\psi(\tau)$ fixes λ , L2 is impossible on that edge: any C1' STATE event in the aligned cell must already be L1. L2 is therefore restricted to misaligned cells in which the contextual transport can move the unprobed pairing into a measured position while preserving the resulting weak order.

The predicted taxonomy was observed in all 225 L2 edges of the primary population. In the 193 T_out/FC_oth edges, exactly one duplicated probed-pairing sum changed and its new value equaled the sum of the unprobed matching; in the 32 three-cycle (Cyc3) cell edges, both duplicated probed-pairing sums changed in the predicted double exchange. The full-intervention $do(c = \gamma)$ block remained pointwise unchanged in all 225 edges. This is an instrument-specific ordinal blindness mechanism; it is distinct from exact pseudometric isometry and should not be interpreted as a new general group-theoretic result.

2.7 Observable-induced indistinguishability shrinks under refinement

The family-20 failure is not an intrinsic absence of identifying information. It emerges along a sequence of representations, several of which are many-to-one. For the frozen contrast, the relevant chain is

$$\begin{aligned} \text{full response pattern} &\rightarrow \text{pointwise popcount} \rightarrow \text{row-resolved field} \rightarrow \text{within-block multiset} \\ &\rightarrow \widetilde{W} \rightarrow d \rightarrow \text{weak rank}. \end{aligned}$$

The $\widetilde{W} \rightarrow d$ step is an exception to the lossy stages in this chain: for the frozen nine-intervention family, the map is injective and admits an exact inverse on its admissible profile image, as shown in Section 2.5.

The L1 route can be localized more precisely. Equality of d_0 and d_1 is already equality of $\widetilde{W}_0$ and $\widetilde{W}_1$; L1 is not produced by cancellation between different measured blocks. Across 22,560 non-null blocks from the studied L1 edges, 0 required cross-block compensation. Exact-profile blindness is therefore resolved locally within the aggregation cells.

The information can disappear at several earlier stages. Among the 698 L1 edges that retained pointwise dependence, 29 (4.2%) became indistinguishable already at pointwise popcount, 174 (24.9%) became indistinguishable when response locations were discarded but within-block multisets were retained, and 495 (70.9%) required strict within-cell arithmetic aggregation: the response multisets differed while their sums agreed. Family 20 lies in this last, dominant mode on both decisive edges.

The same result can be expressed as a hierarchy of blind sets under progressively coarser observables. In the studied 20,000 System-II decisive edges,

$$B_{\text{pattern}} \subset B_{\text{row}} \subset B_{\text{multiset}} \subset B_W,$$

with observed cardinalities

$$7 \subset 36 \subset 210 \subset 705.$$

The nesting expresses the refinement relation; the numerical cardinalities are properties of the frozen generator population, not universal constants. The central implication is that finer observables can strictly recover distinctions lost by coarser ones. The demonstrated blindness is therefore **observable-relative**. It does not imply that the underlying causal system is non-identifiable from all available causal information.

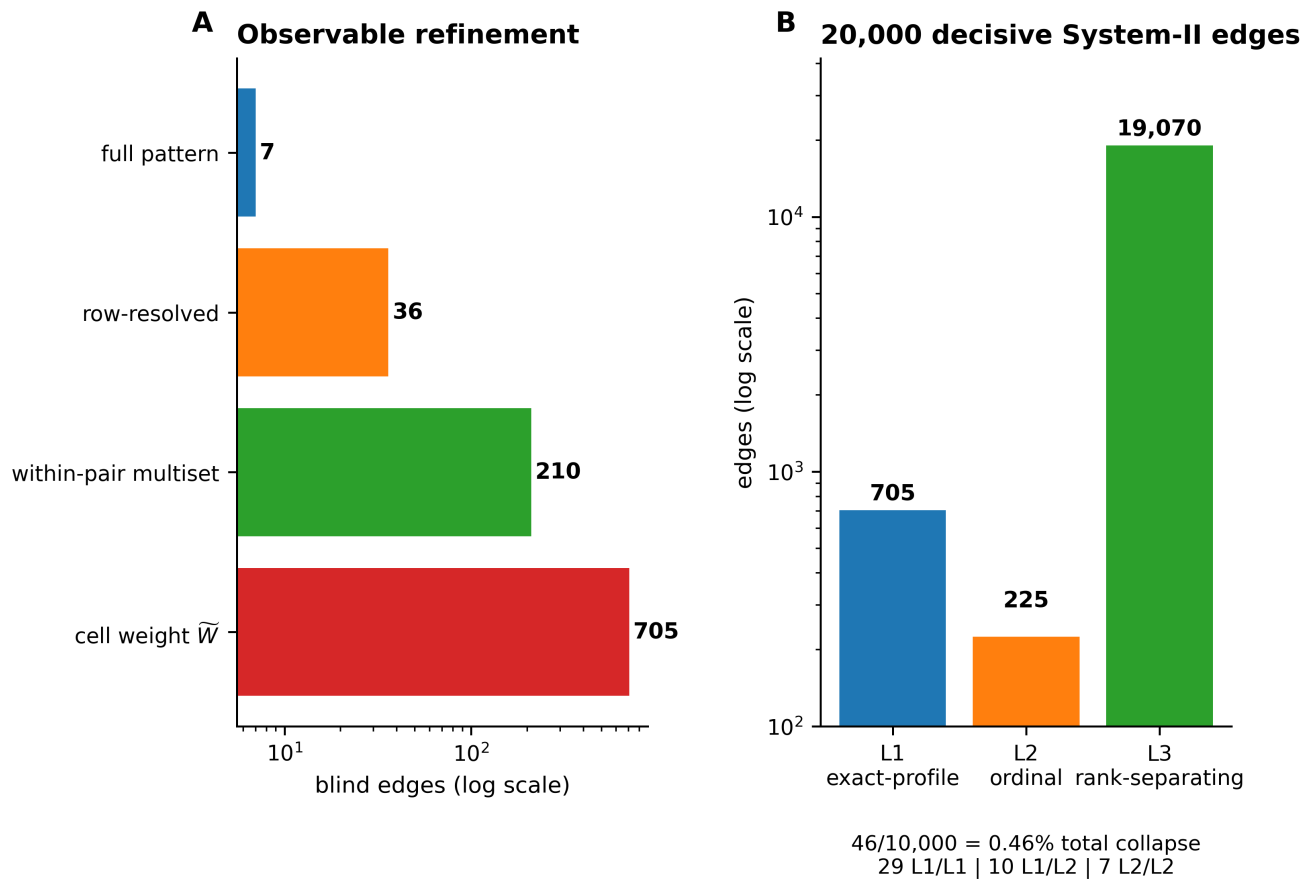


Figure 4. Observable refinement and generator-specific failure structure. Finer response representations strictly reduce the observed indistinguishability set in the frozen generator; at the instance level, total System-II collapse occurs in 46/10,000 eligible families.

2.8 Generator-specific frequency under the frozen generator

To assess whether family 20 recurred within the frozen generator, a post-confirmatory generator-specific frequency study was conducted in which two System-II batches were precommitted before their own generation and analysis. Together they comprised 10,000 eligible families and 20,000 decisive channel-to-processor edges. At the edge level, 705/20,000 (3.53%) were L1, 225/20,000 (1.13%) were L2, and 19,070/20,000 (95.35%) were L3, in which the weak rank changed across receiver-memory contexts. Point-to-point memory dependence was present in 19,993/20,000 edges.

At the instance level, total C1' collapse occurred in 46/10,000 System-II families,

$$0.46\%,$$

with an approximate 95% confidence interval of 0.345%-0.613%. The 46 collapses comprised 29 L1/L1 instances, 10 mixed L1/L2 instances, and 7 L2/L2 instances. Thus exact-profile blindness was the modal route, but ordinal blindness contributed materially to the complete failure class.

Additional precommitted or internally reproduced batches yielded collapse estimates of the same order, approximately 0.32%-0.50% under the frozen generator. These data support recurrence of the mechanism within that generator family; they are not an estimate of a natural or universal frequency, and the internal reproductions do not constitute external scientific replication.

Family 20 is not a separate mechanistic class within this population. It belongs to the modal L1/L1 subtype: one decisive edge has a maximally symmetric equidistant geometry, while the other has a smaller isometry group containing exactly the realized transport. The more detailed correlation analysis between the two decisive edges is reserved for Extended Results/Supplement.

Taken together, the confirmatory counterexample and the population analysis support a bounded conclusion. The preregistered criterion failed. Post-confirmatory analysis showed that one edge-level blind route contributing to that failure is exactly characterized by a receiver-state-dependent relabeling falling inside a symmetry of the receiver’s response pseudometric; a second route arises from ordinal coarsening. In the confirmatory counterexample, the resulting invariance hid a causal difference that remained present in realized local dynamics. The frequency with which these routes occur outside the frozen synthetic generator remains an open empirical question.

3. Discussion

3.1 What failed?

The preregistered criterion failed. C1’ was designed as a conjunctive operational discriminator between two synthetic interaction classes and was evaluated under a blinded confirmatory workflow. It matched the preregistered targets in 150/150 E1 instances and 49/50 E2 instances, but the E2 requirement was conjunctive and therefore failed. The official confirmatory outcome is negative and remains so.

That failure matters more scientifically than the aggregate accuracy. The single C1’ counterexample was not a serialization bug, an inadmissible abstraction, or an instance in which the designed System-II distinction disappeared from the realized dynamics. Post-confirmatory analysis showed that the two decisive receiver-facing edges contained receiver-state-dependent causal differences, including differences strictly realized at the response-sufficient local state (r, c) . Yet C1’ assigned the same STATE relation across contexts and closed the System-II core into the partition associated with the System-III target.

The appropriate interpretation is therefore not that C1’ “almost worked.” In this confirmatory counterexample and for the analyzed one-step receiver-facing contrast, a criterion intended to use interventional invariance as evidence of mechanistic stability encountered a concrete case in which **measured invariance and the independently demonstrated causal difference came apart**.

C2 and C3 also failed their confirmatory targets and are reported as such. They were not subjected to the same mechanistic autopsy and are not retrospectively assigned the mechanism identified for C1’.

3.2 What was characterized?

The post-confirmatory analysis characterized one exact route by which a causal difference can become invisible to the measured representation.

In the frozen System-II contrast, retained receiver state selects a contextual interface relabeling π_m . The two contexts are related by

$$\tau = \pi_1 \circ \pi_0^{-1}.$$

The receiver mechanism itself induces a pairwise response pseudometric W_M . For observables that factor through this geometry, a relative relabeling lying in its isometry group is invisible:

$$\tau \in \text{Iso}(W_M) \implies \widetilde{W}_0 = \widetilde{W}_1.$$

For the frozen nine-intervention family, the measured cardinal profile is an injective measurement representation of $\widetilde{W}$, giving the stronger instrument-specific equivalence

$$d_0 = d_1 \iff \widetilde{W}_0 = \widetilde{W}_1 \iff \tau \in \text{Iso}(W_M).$$

This characterizes the L1 exact-profile blind route. The general mathematics of invariance under symmetry is not new. The specific causal composition is the relevant result: **receiver internal state generates the contextual transformation, the concrete receiver mechanism determines the geometry against which that transformation is symmetric, and the resulting invariance can occur while the underlying causal response remains different and realized**.

This last clause is essential. Symmetry alone establishes blindness of an observable. It becomes causal aliasing only when an underlying causal difference is established independently. Family 20 provides that witness.

The resulting interpretation is therefore representation-relative rather than ontological. In the studied construction,

$$X_0 \neq X_1 \quad \text{while} \quad F(X_0) = F(X_1),$$

for the relevant aggregate F . The causal difference is not absent; it has been quotiented out by the representation retained by the observable.

3.3 What else does the response-order criterion lose?

Exact-profile blindness is not the complete C1' blind set. C1' applies a second coarsening after the cardinal profile d_m : it replaces cardinal values by their weak-order structure.

This creates the L2 route,

$$d_0 \neq d_1, \quad \rho_0 = \rho_1,$$

in which the metric representation detects a contextual change but the final ordinal representation does not. L2 therefore shows that response-order blindness need not require an exact isometry of W_M . Whether any particular L2 edge constitutes causal aliasing additionally requires an independently established underlying causal difference. In the studied instrument, the alignment theorem identifies where this ordinal route is possible: if the contextual transport fixes the unprobed perfect matching, L2 is excluded; the edge can be STATE only through L1. L2 appears only in the misaligned region, where the unprobed pairing can be transported into a measured position while the weak rank is preserved.

More generally, any observable F induces an equivalence relation $X \sim_F X'$ iff $F(X) = F(X')$. The resulting equivalence classes are the most general mathematical object behind observable-induced indistinguishability. A group or stabilizer description is available only when an appropriate group action exists; in the present study $\text{Iso}(W_M)$ characterizes the L1 symmetry route, not the complete blind class of the response-order criterion because L2 is an additional ordinal route.

The information-loss hierarchy gives the same point from a second direction. In the studied 20,000-edge System-II population,

$$B_{\text{pattern}} \subset B_{\text{row}} \subset B_{\text{multiset}} \subset B_W,$$

with cardinalities

$$7 \subset 36 \subset 210 \subset 705.$$

The nesting is structural; the counts are generator-specific. The important conclusion is qualitative: **blindness shrinks when the observable is refined**. What looks invariant after cell aggregation need not be invariant at row, multiset, or point-pattern resolution.

This observation prevents an overly strong interpretation of the negative result. The study does not show that the underlying system is globally non-identifiable from causal information. It shows that identifiability can fail relative to a chosen representation even when a finer representation retains separating information.

3.4 What is not being claimed?

Several broader conclusions are explicitly outside the evidence.

First, STATE and SIGNAL are operational labels of the synthetic generative design. The study does not establish a universal ontology of state coupling and signal-mediated communication, and the family-20 result does not prove that natural systems should be partitioned according to the same labels.

Second, the paper does not establish a universal impossibility of causal individuation. The blind sets are observable-dependent, and finer observables recover distinctions hidden at coarser levels.

Third, the article does not claim a new general theory of symmetry, isometry, quotienting, determining sets, or causal abstraction. Those ideas have substantial prior art. The novelty candidate retained after the adversarial audit is the narrower composition linking receiver internal state, contextual interface relabeling, mechanism-induced response pseudometric, exact blindness, and an independently demonstrated realized causal difference. No equivalent antecedent was identified in that audit; this is not a proof of historical priority.

Fourth, the generator-specific generator-specific frequency,

$$46/10,000 = 0.46\%,$$

is not a universal failure probability. It quantifies recurrence under the frozen eligible generator law only.

Fifth, the temporal result does not establish that causal individuation generally depends on horizon or that increasing temporal depth must destroy specificity. The one-step System-III cancellation is system- and observable-specific, and the horizon-2 3996/4000 observation is empirical under the studied law. This remains a secondary Extended/Supplementary result.

Finally, the article makes no claim about consciousness. Any connection between these causal-individuation questions and theories of consciousness would require a separate conceptual and empirical program.

3.5 Why does it matter?

The practical lesson is methodological. Interventional invariance is often attractive because interventions appear to separate causal structure from mere association. The present result shows that this is not sufficient by itself: **an intervention can be informative at the underlying response-field level while the statistic used to summarize that intervention discards the information that distinguishes the mechanisms.**

This changes the question an analyst should ask. Rather than treating invariance as a property only of the system, one should also ask:

invariance of what representation?

For the studied class, the relevant object is not merely the intervention $do(a)$, but the composition

system response $\rightarrow$ measurement kernel $\rightarrow$ aggregation $\rightarrow$ ordinal decision.

A causal audit can therefore fail not because the intervention had no separating power, but because the downstream observable preserved a quotient in which the causal difference vanished.

This suggests a concrete diagnostic principle for future work: when a high-level interventional criterion reports invariance, test whether the conclusion survives **observable refinement**. In the present study, access to row-resolved, multiset-resolved, or full response-pattern information strictly reduced the blind set. More generally, a finding of invariance should be interpreted relative to the measurement and aggregation pipeline that produced it.

Potential relevance to biological systems and artificial systems

The present study does not establish the mechanism in biology or artificial intelligence, so any domain extrapolation is necessarily prospective. Nevertheless, the result identifies a reason to look for such failures in systems where relevant internal state is partially hidden or where measurements are aggressively coarse-grained.

Complex biological systems are a natural candidate for testing this possibility. They combine latent internal states, feedback, redundancy, multiple organizational scales, and unavoidable experimental coarse-graining. Distinct mechanistic routes can sometimes support similar functional effects, while not all internal variables can be measured or independently manipulated. In such a setting, the practical danger would be to infer mechanistic equivalence from an interventional signature that is invariant only after a particular aggregation. This is a **hypothesis about external relevance**, not evidence that the present blind class is prevalent in biology.

Artificial systems provide a complementary case. Multi-component or multi-agent systems may also contain shared memory, latent state, explicit message channels, and alternative routes for coordination. The difference is that engineered systems can often be instrumented much more deeply: internal activations, memories, network traffic, intermediate states, component ablations, and artificial counterfactual environments may be accessible. In the language of the present results, richer instrumentation can act as an **observable refinement**, potentially moving an audit from a coarse blind representation toward a finer one in which the relevant contextual difference reappears.

That distinction suggests different empirical tests rather than an established ranking. One reason biological systems may provide a particularly stringent test is that experimental access to internal state is often constrained, whereas engineered systems may permit additional measurements that can be used to ask whether a coarse equivalence survives refinement. In high-stakes artificial systems, the consequences of a mistaken certification of

causal separation could also be substantial. Neither comparative generator-specific frequency nor comparative risk was measured here; no ordering between the two domains is a result of this study.

A security-relevant example illustrates the point without claiming that the present generator models such systems. Suppose an audit asks whether two artificial subsystems exchange a message or instead coordinate through shared internal state. If the audit observes only an aggregate interventional signature, two causally different internal implementations could in principle be mapped to the same observed class. The appropriate response is not to assume that communication and shared state are universally indistinguishable, but to test whether additional instrumentation changes the conclusion—for example by inspecting intermediate state, blocking specific channels, cloning internal states, or measuring additional response features. The present hierarchy shows that refinement can shrink a blind set in the studied class; it does not guarantee that any particular finite refinement will resolve ambiguity in an arbitrary real-world system.

A distinct speculative future problem: intervention-induced compensatory aliasing

The mechanism characterized in this paper is **representation-induced**. The intervention produces a causal response difference, but the downstream representation identifies the two contexts.

A separate phenomenon may arise in adaptive systems if the intervention changes the mechanism that generates the response. Feedback, homeostasis, or redundant pathways could in principle produce a compensatory reconfiguration after intervention, causing macroscopically similar responses even though the internal causal organization has changed. The phrase **intervention-induced compensatory aliasing** is used here only as a provisional future-work label; neither the mechanism nor the terminology is an empirical result or novelty claim of the present study.

That hypothetical process should not be conflated with the mechanism demonstrated here. The distinction is:

representation-induced aliasing: $X_0 \neq X_1$ but $F(X_0) = F(X_1)$,

versus a possible

intervention-induced compensatory aliasing: $do(a)$ changes the operative dynamics $T \rightarrow T_a$,

after which adaptation or compensation drives the measured responses toward equivalence.

The present study provides no evidence that this second phenomenon occurred in the frozen systems. It is proposed only as a future question: **can observational blindness caused by representation be formally distinguished from response equivalence actively produced by the system’s adaptation to intervention?**

That question may be particularly relevant to biological regulation and adaptive artificial systems, where interventions can themselves alter the internal regime being interrogated.

Broader programmatic interpretation

Taken together, the results motivate—but do not prove—a broader programmatic view in which observed causal individuation can depend jointly on the system, the observable, and the temporal horizon through which the system is interrogated.

The evidence in this paper directly establishes observable dependence only within the studied construction and treats the temporal component as a separate, narrow result. Therefore,

$$\text{observed individuation} = f(\text{system, observable, horizon})$$

should be read as a hypothesis for future research, not as a theorem of the present study.

The immediate contribution is narrower and testable: **measured causal invariance need not imply causal constancy when contextual causal differences lie in a blind class of the representation used to measure them.**

4. Methods

Nomenclature and traceability

The manuscript uses descriptive scientific terms in the main prose while retaining the frozen preregistration identifiers for auditability.

Manuscript-facing term	Frozen identifier	Function
interventional response-order invariance criterion	C1'	scientific candidate; compares weak order of response profiles across receiver-memory contexts
effective-intervention partial-order invariance criterion	C2	scientific candidate; compares inclusion order of effective intervention sets
support-invariance implementation canary	C3	implementation canary; does not enter H_A
direct-coupling benchmark	System I	benchmark without explicit interface modules
receiver-state-remapped interface system	System II	critical target class with memory-dependent interface relabeling
direct-interface benchmark	System III	interface read directly, without contextual relabeling
base stratum	E1	10-bit core systems
downstream-extension stratum	E2	core plus two downstream no-feedback bits
exact-profile blind route	L1	$d_0 = d_1$
ordinal blind route	L2	$d_0 \neq d_1$ but equal canonical weak rank
rank-separating case	L3	canonical weak rank differs

4.1 Operational targets and construct validity

Synthetic systems and module structure

The study used deterministic finite-state systems designed as a controlled testbed for an operational distinction between two forms of interaction. The labels **STATE** and **SIGNAL** are therefore **classifier labels and preregistered design targets**, not claims that the synthetic systems instantiate a universal ontology of physical, biological, or communicative interaction.

The two processor modules were

$$A = (x, m_A), \quad B = (y, m_B),$$

where $x, y \in \{0, 1, 2, 3\}$ are two-bit processor states and $m_A, m_B \in \{0, 1\}$ are retained one-bit receiver states. Systems II and III additionally contained two two-bit interface modules,

$$C_{AB} = c_{AB}, \quad C_{BA} = c_{BA},$$

with $c_{AB}, c_{BA} \in \{0, 1, 2, 3\}$. Thus System I had six state bits and Systems II/III had ten. The second confirmatory stratum (E2) appended two one-bit downstream modules D_1, D_2 , giving twelve bits. The downstream modules read the previous core state but did not feed back into the core:

$$D_{i,t+1} = R_i(s_t^{\text{core}}).$$

For each sampled family θ , the common processor and memory maps were

$$F_0, G_0 : \{0, 1, 2, 3\}^2 \rightarrow \{0, 1, 2, 3\},$$

$$H, K : \{0, 1\} \times \{0, 1, 2, 3\} \rightarrow \{0, 1\},$$

with memory-dependent additive terms $\sigma_A, \sigma_B : \{0, 1\} \rightarrow \{0, 1, 2, 3\}$. System II additionally used two distinct permutations $\pi_0, \pi_1 \in S_4$.

Dynamics

System I implemented direct reciprocal processor input:

$$u_t = y_t, \quad v_t = x_t,$$

$$x_{t+1} = F_0(x_t, u_t) \oplus \sigma_A(m_{A,t}), \quad m_{A,t+1} = H(m_{A,t}, x_t),$$

$$y_{t+1} = G_0(y_t, v_t) \oplus \sigma_B(m_{B,t}), \quad m_{B,t+1} = K(m_{B,t}, y_t).$$

Systems II and III used explicit interface variables, with

$$c_{AB,t+1} = x_t, \quad c_{BA,t+1} = y_t.$$

In System III, the processors read the interface directly,

$$v_t = c_{AB,t}, \quad u_t = c_{BA,t}.$$

In System II, receiver memory contextually relabeled the incoming interface:

$$v_t = \pi_{m_{B,t}}(c_{AB,t}), \quad u_t = \pi_{m_{A,t}}(c_{BA,t}).$$

The processor and memory update equations otherwise remained the same across the three systems.

Preregistered operational partitions

The preregistered scoring target was defined at the level of the component partition induced by the classifier and, for unprojected systems, its corresponding set of causal media. The target was fixed by construction:

Stratum / system	Target component partition	Target media
E1 / I	$\{A, B\}$	$\{A, B\}$
E1 / II	$\{A\}, \{B\}, \{C_{AB}\}, \{C_{BA}\}$	$\{A\}, \{B\}$
E1 / III	$\{A, B, C_{AB}, C_{BA}\}$	$\{A, B, C_{AB}, C_{BA}\}$
E2 / II	$\{A\}, \{B\}, \{C_{AB}\}, \{C_{BA}\}, \{D_1\}, \{D_2\}$	$\{A\}, \{B\}$
E2 / III	$\{A, B, C_{AB}, C_{BA}\}, \{D_1\}, \{D_2\}$	$\{A, B, C_{AB}, C_{BA}\}$

These partitions are **pre-specified targets of the synthetic generative design**. The study asks whether a frozen interventional classifier preserves that designed mechanistic distinction under blinding, scale projections, and admissible representational equivalences. It does not establish that the labels System II/System III, STATE/SIGNAL, or the resulting partitions are a universal taxonomy of interactions in natural systems.

4.2 Preregistered protocol, generation, and blinding

Family generation

A family θ comprised $F_0, G_0, H, K, \sigma_A, \sigma_B, \pi_0, \pi_1$, initial processor/memory/interface states, and, in E2, downstream lookup tables R_1, R_2 and initial downstream bits. The base tables were sampled uniformly over their finite codomains. Initial processor, memory, and interface states were sampled uniformly over their finite state spaces; in E2, the initial downstream bits were also sampled uniformly. The two interface permutations were sampled independently and families with $\pi_0 = \pi_1$ were rejected.

Randomness was a deterministic function of the root seed using NumPy PCG64. Four child streams were assigned positionally to (i) nuclear family generation, (ii) export permutations, (iii) instance shuffling/identifiers, and (iv) E2 downstream extensions. E2 extensions were sampled only after the nuclear family passed eligibility, preventing rejected nuclear families from consuming downstream random draws.

Eligibility was assessed on the core dynamics before blind export. For every accepted family, Systems I, II, and III had to satisfy the preregistered conditions E1-E5, including visitation of both retained-memory contexts, efficacy of both incoming processor inputs, recurrence witnesses linking processor state and retained memory, sufficient local-state reachability, and non-degenerate memory-dependent additive effects. System II additionally had to satisfy E6, a witnessed memory-dependent remapping effect in both directions. Downstream E2 modules were excluded from eligibility because they had no feedback into the core.

Confirmatory strata

The first confirmatory stratum E1 contained 50 accepted families. Each family generated one instance of System I, one of System II, and one of System III, for

$$50 \times 3 = 150$$

instances.

The second stratum E2 contained 25 accepted extended families. Each generated one System-II and one System-III instance with downstream modules D_1, D_2 , for

$$25 \times 2 = 50$$

instances.

The confirmatory root seeds were fixed before generation and each was used exactly once. Their values and the complete execution manifests are retained in the reproducibility record described in Section 4.15.

Blind export

Before analysis, each instance underwent two separate randomizations for label destruction. First, the state-bit coordinates were randomly permuted. Second, module order was randomly permuted. Module boundaries and retained-memory bit designations remained visible because they were required by the classifier, but semantic module identities were replaced by anonymous identifiers $Q_0, Q_1, \dots$. No marker identified system type, family membership, or paired variants.

For each instance, the hidden key recorded the system variant, bit permutation, module-order permutation, family index, and a hash of the generating θ . The key was serialized separately from the blinded transition table. Blinding was procedural rather than cryptographic: code and seed could in principle regenerate the key. The frozen procedure therefore required that the key be moved outside the classifier-accessible directory before analysis, that the generator not be rerun with a confirmatory seed during analysis, and that the key be opened only at the prespecified scoring step after verification against its recorded SHA-256 digest.

The classifier itself was a pure map from a blinded instance to a classification and contained no key-reading path. Confirmatory output directories were required to be new, canonical JSON bytes were hashed at write time, and the post-confirmatory workflow left the official confirmatory artifacts unchanged.

4.3 Interventions, one-step interventional influence graph, response-order criterion, and confirmatory scoring

One-step intervention semantics

For a module A with coordinate set B_A , the intervention family was complete over coordinate rewrites:

$$\mathcal{I}_A = \left\{ do(J = \eta) : J \subseteq B_A, \eta \in \{0, 1\}^{|J|} \right\}.$$

The null intervention $do(\emptyset)$ was included. Hence

$$|\mathcal{I}_A| = 3^{|B_A|}.$$

Interventions were transient: the selected sender bits were overwritten and the system then underwent **one** global transition. For a two-bit channel this produced nine interventions, in the frozen order: null; low bit set to 0 or 1; high bit set to 0 or 1; and the full channel set to each of 0, 1, 2, 3.

One-step interventional influence graph G_C

Let $\mathcal{O}(s_0)$ be the deterministic orbit from the initial state to first repetition. For distinct modules $A \neq B$, an edge $A \rightarrow B$ was placed in the pre-classification one-step interventional influence graph G_C when there existed a non-null intervention on A and an orbit-reached context for which the next state of B differed from the null-intervention next state. Self-edges were excluded.

Thus the subsequent STATE/SIGNAL decision was made only for edges with a witnessed one-step causal effect in G_C .

Counterfactual fibers and the response-order criterion (C1')

For a receiver B with retained-memory coordinates M_B , let $\mathcal{M}_B^{\text{reach}}$ denote the memory values reached on the orbit. For each reached value m , the counterfactual fiber contained **all** global states with receiver memory fixed to m . All remaining coordinates formed an aligned free axis r , identical across memory contexts.

For an intervention $a \in \mathcal{I}_A$, define the receiver response profile

$$d_m(a) = \sum_r \text{Ham}_B(\Phi^a(r, m), \Phi^0(r, m)),$$

where $\Phi^a(r, m)$ is the next receiver state after applying intervention a to the global state indexed by (r, m) , and Φ^0 is the null-intervention baseline.

C1' does not require equality of the cardinal values $d_m(a)$. It compares their **weak order with ties**. The canonical weak rank of profile entry a is

$$\rho_m(a) = |\{d_m(a') : d_m(a') < d_m(a)\}|,$$

where duplicate smaller values count once. Two profiles have the same canonical rank vector if and only if they have the same weak order and the same tie structure.

A causal edge $A \rightarrow B$ was labeled

$$\text{STATE} \iff \rho_m = \rho_{m'} \text{ for all reached } m, m',$$

and SIGNAL otherwise.

For a receiver with no retained-memory coordinates, the frozen classifier used the degenerate context set

$$\mathcal{M}_B^{\text{reach}} = \{\perp\}.$$

With only one context, cross-context invariance is vacuous. Consequently, every causal edge into a memoryless receiver is labeled STATE by C1', C2, and C3. This rule is essential to reproducing the System-II STATE ring: in that architecture the memoryless channel receivers contribute STATE edges automatically, so a total C1' collapse is determined by the two decisive channel-to-processor edges.

C2 and C3

C2 used a different invariant. For each receiver output bit b_k , it formed the set of interventions on the sender that were effective on that bit within a fixed memory context. C2 labeled an edge STATE when the complete partial-order relation of inclusion and equality among these effective-intervention sets was invariant across reached receiver-memory contexts.

C3 was the prespecified support-invariance implementation canary. For each receiver output bit it recorded whether the exact intervention-response signatures varied across interventions, producing a support set for each memory context. C3 labeled an edge STATE when this support set was invariant. C3 was executed and scored for audit purposes but was not a scientific H_A decision candidate.

STATE graph, components, and media

For each candidate $C \in \{C1', C2, C3\}$, the STATE graph G_S^C retained only those edges of G_C labeled STATE by C . The component partition was the set of strongly connected components of G_S^C .

For unprojected confirmatory systems, a causal medium was either (i) a strongly connected component containing at least two modules, or (ii) a singleton module carrying preregistered retained memory. For accepted unprojected instances, the recurrence requirement of such memory-bearing singletons had already been certified by eligibility and was not re-inferred from the net transition-table effect, because cancellation between σ and π_m can make that net effect misleading in System II. Modules without retained memory could not form singleton media.

Scale-preservation item

The confirmatory score also required stability under admissible coordinate projections in **both E1 and E2**. For every subset S with

$$2 \leq |S| < n,$$

the scale executor first tested exact fiber consistency of the projected transition for **every intervention on subsets of S** . If the projection was admissible, it constructed the quotient transition T_S , projected the initial state and surviving modules, projected retained-memory coordinates by intersection with S , and applied the same frozen classifier. For each candidate, scale preservation required that every pair of surviving modules be in the same component after projection if and only if it was in the same component in the micro system. Projected media could be constructed by the classifier but were not a separate scored scale item; the scale score compared the component relation among surviving modules.

In E2, execution validity additionally required at least two admissible projections of distinct granularities. Failure of this construction requirement annulled the execution rather than defeating a candidate.

Representational-equivalence item

The final scientific invariance item exhaustively enumerated the preregistered group of coordinate representations that preserve modular form, memory designation, and the atomic intervention algebra:

1. module renamings only within equal module forms ($n_{\text{bits}}, n_{\text{memory bits}}$);
2. within-module coordinate permutations with memory coordinates mapped to memory coordinates and non-memory coordinates to non-memory coordinates;
3. independent bit flips.

The transition table and initial condition were conjugated under each transformation. Arbitrary S_{2^d} state bijections were excluded because they do not preserve coordinate interventions. For every transformed instance, G_C , edge labels, component partitions, and media were compared after the corresponding module renaming. A discrepancy in the conjugated G_C invalidated the representational-equivalence audit procedurally rather than being counted as an ordinary scientific candidate mismatch. Confirmatory enumeration was exhaustive; the expected group sizes were 512 for E1/System I, 65,536 for E1/Systems II and III, and 524,288 for E2/Systems II and III.

Candidate and experiment-level decision rules

Before any scientific candidate was evaluated, the scoring program audited exact instance cardinalities and IDs, exhaustive equivalence counts, E2 scale validity, and the C3-specific canary condition. The canary was evaluated on the 75 System-II confirmatory instances - 50 in E1 and 25 in E2 - as a procedural implementation check distinct from scientific candidate success. Integrity, E2-construction, or canary failures produced procedural annulment states rather than scientific candidate defeats.

For each of C1', C2, and C3, a candidate-level pass required the conjunction of four items:

$$\text{targets}_{E1} \wedge \text{targets}_{E2} \wedge \text{scale} \wedge \text{equivalences}.$$

All 200 micro-level confirmatory instances counted toward the target items; a wrong partition that was stable under scale or equivalence transformations did not count as success.

The scientific confirmatory result H_A was positive if and only if C1' **or** C2 passed all four items. C3 did not enter H_A . The frozen executable contained no C4 scoring path.

4.4 Response-field definitions for the post-confirmatory analysis

The post-confirmatory analysis retained the frozen intervention semantics and classifier state fibers but made the pointwise response field explicit.

For a causal edge from sender A to receiver B , fix a reached receiver-memory context m . Let Z_m be the full aligned fiber of global states whose receiver-memory coordinates equal m . For each free-axis index r , write $z(r, m) \in Z_m$.

Let e_B extract the receiver module from a global state and let I_a denote the overwrite map for intervention a . Define

$$\Phi_m^a(r) = e_B[T(I_a(z(r, m)))],$$

and the pointwise intervention response relative to the null intervention as

$$X_m^a(r) = \Phi_m^a(r) \oplus \Phi_m^0(r).$$

The frozen C1' cardinal profile is therefore

$$d_m(a) = \sum_r \text{pc}(X_m^a(r)).$$

For the post-confirmatory cross-context analysis, pointwise memory dependence was defined by

$$X_0^a(r) \neq X_1^a(r).$$

The total pointwise dependence count for an edge was the number of intervention/free-axis pairs (a, r) satisfying this inequality. This quantity is finer than C1' because C1' first converts the response pattern to Hamming weight and then aggregates over the full fiber before applying weak rank.

The post-confirmatory levels used throughout the analysis were:

$$\text{L1 : } d_0 = d_1,$$

$$\text{L2 : } d_0 \neq d_1 \wedge \rho_0 = \rho_1,$$

$$\text{L3 : } \rho_0 \neq \rho_1.$$

L1 and L2 are therefore distinct routes to the same C1' STATE edge label; L3 is a SIGNAL edge.

4.5 Response-sufficient local-state comparability and realized differences

Post-confirmatory analysis. Sections 4.5-4.9 were developed after confirmatory scoring and do not alter the frozen confirmatory criterion or result.

The full counterfactual fiber deliberately ranges over global states that need not all be visited by the orbit. To test whether a cross-context causal difference survived in realized dynamics, realized support was defined separately.

For a channel-to-processor edge in System II, let r denote the non-memory processor state of the receiver and let c denote the incoming channel symbol. For memory context m , define

$$\mathcal{C}_m = \{(r, c) : (r, c, m) \text{ occurs in a state on the deterministic orbit}\}.$$

The strict realized support is

$$\mathcal{I} = \mathcal{C}_0 \cap \mathcal{C}_1.$$

Local sufficiency

For the frozen one-step System-II channel-to-processor contrast, conditional on receiver memory m , the pointwise response to a channel intervention is a function only of (r, c) . All other global coordinates are irrelevant to that response at the current step. Consequently, (r, c) is a response-sufficient comparability state for this contrast. No minimality claim is required or made:

- refining (r, c) with additional global coordinates can only reduce the observed support intersection by demanding equality of coordinates that do not affect the response;
- coarsening (r, c) may merge states that have different one-step responses and therefore need not preserve the response comparison.

A **strictly realized dependency site** was therefore an intervention/cell pair

$$(a, (r, c)), \quad (r, c) \in \mathcal{I},$$

for which

$$X_0^a(r, c) \neq X_1^a(r, c).$$

This definition does not require the complete global state to be identical across contexts; it requires equality of the local state that is sufficient for the causal contrast being tested.

Realized-transition observational witness

As a complementary check requiring no counterfactual overwrite, for two channel values c_1, c_2 realized with the same receiver state r under both memory contexts, the analysis defined

$$O_m(r; c_1, c_2) = e_B[T(s(r, c_1, m))] \oplus e_B[T(s(r, c_2, m))].$$

A realized-transition witness exists when

$$O_0(r; c_1, c_2) \neq O_1(r; c_1, c_2).$$

This test asks whether the context-dependent interface effect can be seen directly in transitions that the deterministic trajectory actually visits.

4.6 Contextual pull-back lemma for System II

Post-confirmatory analytic characterization.

Consider a decisive System-II channel-to-processor edge and condition on receiver memory m .

For $C_{AB} \rightarrow B$, write the receiver processor state as $r = y$, the incoming channel value as $c = c_{AB}$, and $M = G_0$. The next processor state is

$$M[r][\pi_m(c)] \oplus \sigma[m].$$

The receiver memory update $K[m][r]$ is independent of the channel value c . Thus a channel intervention changes neither the current-step receiver-memory update nor any other component of the receiver module except through the processor response.

For $C_{BA} \rightarrow A$, the same form holds with $r = x$, $c = c_{BA}$, $M = F_0$, and the receiver-memory update $H[m][r]$.

Let $\text{sub}_a(c)$ denote the channel symbol obtained after applying the bit-rewrite intervention a . The XOR intervention response is

$$\begin{aligned} X_m(a; r, c) &= (M[r][\pi_m(\text{sub}_a(c))] \oplus \sigma[m]) \\ &\quad \oplus (M[r][\pi_m(c)] \oplus \sigma[m]) \\ &= M[r][\pi_m(\text{sub}_a(c))] \oplus M[r][\pi_m(c)]. \end{aligned}$$

Hence $\sigma[m]$ cancels exactly. Receiver memory remains causal in this contrast through its selection of π_m , not through the additive term.

Define the context-free response tensor

$$X_\star(a; r, c) = M[r][\text{sub}_a(c)] \oplus M[r][c].$$

Then the context-specific field is the pull-back of the same receiver mechanism by π_m : memory changes which interface symbol is presented to M , while the receiver table itself is fixed. The relative contextual transport is

$$\tau = \pi_1 \circ \pi_0^{-1}.$$

This reduction is specific to the proven one-step comparison class: aligned full fibers, contextual interface relabeling, channel-independent current-step memory update, and XOR differencing. It is not assumed for arbitrary dynamics or arbitrary temporal horizons.

4.7 Mechanism-induced response pseudometric

Post-confirmatory analytic characterization.

For a fixed receiver table M , define the row-resolved pairwise response weight

$$w_r(p, q) = \text{pc}_2(M[r][p] \oplus M[r][q]),$$

where pc_2 is Hamming popcount on the two processor-output bits. The aggregate receiver-response pseudometric is

$$W_M(p, q) = \sum_{r=0}^3 w_r(p, q) = \sum_{r=0}^3 \text{pc}_2(M[r][p] \oplus M[r][q]).$$

W_M is symmetric with zero diagonal and depends on the concrete receiver mechanism M .

The memory-context pull-back of this geometry is

$$\widetilde{W}_m(c_1, c_2) = W_M(\pi_m(c_1), \pi_m(c_2)).$$

Using the lemma in Section 4.6 and summing over the aligned full fiber, the frozen C1' cardinal statistic on a channel intervention a has the exact form

$$d_m(a) = 2^{n-5} \sum_{c=0}^3 \widetilde{W}_m(c, \text{sub}_a(c)).$$

The factor 2^{n-5} is a context-independent multiplicity contributed by the remaining free coordinates of the full fiber. Thus the same core statistic is multiplied by 32 for $n = 10$ and by 128 for $n = 12$. In particular, the two downstream E2 bits multiply the corresponding E1 channel statistic by four without changing its weak order.

Define

$$\text{Iso}(W_M) = \{g \in S_4 : W_M(gp, gq) = W_M(p, q) \ \forall p, q\}.$$

This is the group of interface permutations invisible to the aggregate receiver-response pseudometric.

4.8 Class-level symmetry-blindness proposition

Post-confirmatory analytic characterization.

The mechanism above can be stated without reference to the particular generator.

Let C be a finite interface alphabet and suppose two contexts $m \in \{0, 1\}$ act on a fixed response mechanism only by interface relabelings π_m . Let W be the pairwise response pseudometric induced by that fixed mechanism, and define

$$\widetilde{W}_m(c_1, c_2) = W(\pi_m(c_1), \pi_m(c_2)).$$

Let

$$\tau = \pi_1 \circ \pi_0^{-1}.$$

Consider any observable F whose contextual input factors only through the cell-weight field $\widetilde{W}_m$. If

$$\tau \in \text{Iso}(W),$$

then for every pair c_1, c_2 ,

$$\begin{aligned} \widetilde{W}_1(c_1, c_2) &= W(\pi_1(c_1), \pi_1(c_2)) \\ &= W(\tau\pi_0(c_1), \tau\pi_0(c_2)) \\ &= W(\pi_0(c_1), \pi_0(c_2)) \\ &= \widetilde{W}_0(c_1, c_2). \end{aligned}$$

Therefore,

$$\boxed{\tau \in \text{Iso}(W) \implies F(\widetilde{W}_0) = F(\widetilde{W}_1)}.$$

This proposition establishes **blindness of the observable class**, not by itself a causal difference between contexts. The invariance is a case of standard symmetry/invariance mathematics. It becomes causal aliasing only when a difference in the underlying response fields is independently established, as done for the confirmatory counterexample in Section 2.2.

The proposition is deliberately restricted to observables that factor through the cell-weight field. Row-resolved, multiset-resolved, or full-pattern observables need not share the same blind set.

4.9 Exact characterization of the exact-profile blind route

Post-confirmatory analytic characterization.

The frozen two-bit channel intervention set makes the cell-weight condition exact, but the underlying identification condition can be stated for an arbitrary linear intervention summary. Let $\mathcal{A}_I$ act on symmetric zero-diagonal pairwise arrays and write $d_m = \mathcal{A}_I(\widetilde{W}_m)$. Then

$$d_0 = d_1 \iff \widetilde{W}_1 - \widetilde{W}_0 \in \ker(\mathcal{A}_I).$$

Consequently, if

$$\ker(\mathcal{A}_I) \cap \Delta\mathcal{O}(W_M) = \{0\},$$

where $\Delta\mathcal{O}(W_M)$ is the difference set of the relabeling orbit of W_M , the operator separates every contextual relabeling relevant to that mechanism and the exact converse to Proposition 1 holds. Global injectivity on the full pairwise-array space is sufficient but not necessary; orbit-separation is the sharper condition.

For the frozen four-symbol instrument the ambient dimension is six. Let the six independent off-diagonal entries of a context-specific geometry $\widetilde{W}$ be

$$\mathbf{w} = (w_{01}, w_{02}, w_{03}, w_{12}, w_{13}, w_{23})^\top.$$

The nine channel interventions occur in the order

$$\emptyset, b_0:=0, b_0:=1, b_1:=0, b_1:=1, c:=0, c:=1, c:=2, c:=3.$$

After division by the common multiplicity 2^{n-5} , their aggregate profile has the form

$$(0, A, A, B, B, V_0, V_1, V_2, V_3),$$

where

$$A = w_{01} + w_{23}, \quad B = w_{02} + w_{13},$$

and

$$\begin{aligned} V_0 &= w_{01} + w_{02} + w_{03}, \\ V_1 &= w_{01} + w_{12} + w_{13}, \\ V_2 &= w_{02} + w_{12} + w_{23}, \\ V_3 &= w_{03} + w_{13} + w_{23}. \end{aligned}$$

Thus the six measured quantities $\mathbf{s} = (A, B, V_0, V_1, V_2, V_3)^\top$ are related to $\mathbf{w}$ by

$$\mathbf{s} = L\mathbf{w},$$

with

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

Direct calculation gives

$$\text{rank}(L) = 6, \quad \det L = -8.$$

An exact integer-rank enumeration over all 512 subsets of the nine intervention rows gives the following number of rank-6 subsets by subset size:

$$N_6 = 4, \quad N_7 = 8, \quad N_8 = 5, \quad N_9 = 1,$$

with no rank-6 subset of size smaller than six. The four minimal subsets all contain $c := 0, 1, 2, 3$, exactly one of $b_0 := 0$ or $b_0 := 1$, and exactly one of $b_1 := 0$ or $b_1 := 1$. The null intervention is never required for rank. This audit is algebraic and post-confirmatory; it uses only the frozen intervention measurement rows.

Hence L is invertible. Equality of the complete frozen cardinal profiles is therefore equivalent to equality of the context-specific cell-weight geometries:

$$d_0 = d_1 \iff \widetilde{W}_0 = \widetilde{W}_1.$$

The converse is non-trivial for this instrument because the nine interventions directly probe only two of the three perfect matchings on four channel symbols. The third matching is recovered indirectly through the four degree sums $V_0, \dots, V_3$. The two restricted intervention subfamilies examined in the analysis have ranks 2 and 4, respectively, and do not determine all six entries of $\widetilde{W}$.

Finally, because

$$\widetilde{W}_m(c_1, c_2) = W_M(\pi_m(c_1), \pi_m(c_2)),$$

it follows that

$$\widetilde{W}_0 = \widetilde{W}_1 \iff \pi_1 \pi_0^{-1} \in \text{Iso}(W_M).$$

Combining the two equivalences yields the exact characterization of the frozen **L1 exact-profile blind route**:

$$\boxed{d_0 = d_1 \iff \widetilde{W}_0 = \widetilde{W}_1 \iff \tau \in \text{Iso}(W_M)}.$$

This theorem is specific to L1. C1' subsequently coarsens d_m to its weak-order representation, so equality of weak ranks can also occur when $d_0 \neq d_1$; that separate L2 ordinal route is treated in Section 4.10.

The invertibility of L is a full-rank measurement property of the frozen intervention design and is not presented as a new general mathematical principle.

4.10 Ordinal blindness and the alignment theorem

L1 and L2 are distinct routes to the same C1' edge label

For the decisive two-bit channel-to-processor edges, C1' assigns the edge label STATE when the canonical weak-rank vectors of the two receiver-memory contexts agree. Consequently, the post-confirmatory analysis separated two cases:

$$\text{L1 : } d_0 = d_1,$$

and

$$\text{L2 : } d_0 \neq d_1 \quad \wedge \quad \rho_0 = \rho_1.$$

Both L1 and L2 therefore produce a STATE edge under C1', but only L1 is characterized by the exact isometry result of Section 4.9. L2 is an additional information-loss stage created by the final cardinal-to-ordinal coarsening.

For a two-bit channel, after removal of the common multiplicity, the frozen profile has the structure

$$(0, P_1, P_1, P_2, P_2, D_0, D_1, D_2, D_3),$$

where P_1 and P_2 are the sums over the two perfect matchings directly probed by the single-bit interventions and $D_0, \dots, D_3$ are the four degree sums generated by the full-channel interventions.

Action on the three perfect matchings

The complete graph on the four interface symbols $C = \{0, 1, 2, 3\}$ has three perfect matchings:

$$\lambda_1 = \{\{0, 1\}, \{2, 3\}\},$$

$$\lambda_2 = \{\{0, 2\}, \{1, 3\}\},$$

$$\lambda_3 = \{\{0, 3\}, \{1, 2\}\}.$$

The frozen single-bit interventions directly probe two of these pairing sums. Let λ denote the remaining, unprobed matching in the relevant coordinate representation.

Any interface permutation $g \in S_4$ acts on the set of the three perfect matchings. Let

$$\psi : S_4 \rightarrow S_3$$

denote this action. Its kernel is the Klein four group

$$V_4 = \{\text{id}, (01)(23), (02)(13), (03)(12)\}.$$

The relative receiver-state transport

$$\tau = \pi_1 \circ \pi_0^{-1}$$

therefore induces a permutation $\psi(\tau)$ of the three matching sums.

Alignment theorem

The post-confirmatory alignment theorem concerns an **edge-level** C1' decision.

If

$$\psi(\tau)(\lambda) = \lambda,$$

then the contextual transport does not move the unprobed matching into either of the two directly probed pairing positions. Under this condition, the context-1 frozen profile is obtained from the context-0 profile only by the corresponding positional permutation of quantities already present in the measured profile.

Because C1' requires the canonical weak rank to remain fixed at the same intervention positions, an aligned edge cannot enter L2. More precisely,

$$\boxed{\psi(\tau)(\lambda) = \lambda \implies (\rho_0 = \rho_1) \Rightarrow d_0 = d_1}.$$

Thus, in an aligned cell, every C1' STATE event on the decisive edge is L1. L2 is possible only when the contextual transport moves the unprobed matching into a probed position.

This result is instrument-specific. The group action $S_4 \rightarrow S_3$ and its kernel V_4 are classical; the relevant result here is the consequence of that action for the frozen intervention profile.

Value mechanism in the L2 region

When $\psi(\tau)$ does not fix λ , the unprobed matching can be transported into the measured P block. For transposition/4-cycle misaligned cells (T_out and FC_oth), one directly probed matching sum is replaced by the previously unprobed sum. For the three-cycle (Cyc3) cell, the induced action cyclically transports the matching sums and both measured P blocks undergo the predicted double exchange. Let

$$S(\lambda) = \sum_{\{p,q\} \in \lambda} W_M(p, q).$$

In T_out and FC_oth L2 cases, the changed duplicated pairing block receives the value transported from $S(\lambda)$. In Cyc3 L2 cases, both duplicated pairing blocks change under the corresponding cyclic transport. In all cases, the full-channel $do(c = \gamma)$ degree block remains pointwise unchanged. The cardinal profile can therefore change without changing its weak order.

This taxonomy was checked against the complete primary L2 population. Among the 225 L2 edges, 193 were T_out/FC_oth and changed exactly one duplicated P block; in all 193, the new value equaled the predicted $S(\lambda)$. The remaining 32 were Cyc3-cell edges and changed both duplicated P blocks in the predicted double exchange. The full-channel degree block was unchanged in 225/225 edges. The aligned-region exclusion was additionally tested on 21,390 aligned edges accumulated across the primary, precommitted OOS, and RAW analyses, with zero L2 counterexamples.

The L2 mechanism is not an exact-isometry mechanism in disguise. In the primary set of 46 total-collapse instances, 17 of the 24 L2 edges had trivial $\text{Iso}(W_M)$. Their metric profiles changed; only the final weak-order representation remained invariant.

4.11 Strict-realization analysis and witness extraction

All analyses in this section are post-confirmatory. They were designed to determine whether the response differences hidden by the aggregate occurred in states that were actually visited under both receiver-memory contexts.

Realized local support

For a decisive System-II channel-to-processor edge, let $r \in \{0, 1, 2, 3\}$ be the non-memory processor state of the receiver and let $c \in \{0, 1, 2, 3\}$ be the incoming channel symbol. For receiver-memory context m , define the realized support

$$\mathcal{C}_m = \{(r, c) : \exists s \in \mathcal{O}(s_0) \text{ with local receiver/channel state } (r, c, m)\},$$

where $\mathcal{O}(s_0)$ is the deterministic orbit from the initial condition to first repetition.

The common realized support is

$$\mathcal{I} = \mathcal{C}_0 \cap \mathcal{C}_1.$$

The analysis used **support**, not visitation frequency: a local state counted as realized if it occurred at least once in the orbit under the relevant memory context.

Why (r, c) is the comparison state

For the frozen one-step System-II channel-to-processor contrast, conditional on memory m , the intervention response is

$$X_m(a; r, c) = M[r][\pi_m(\text{sub}_a(c))] \oplus M[r][\pi_m(c)].$$

No other global coordinate appears in this expression. Therefore (r, c) is a response-sufficient local state for this contrast.

This choice was not made to maximize overlap after observing the failed instance. It follows directly from the frozen transition law. Refining the comparison state with globally irrelevant coordinates can make two response-equivalent local situations appear incomparable; coarsening below (r, c) may merge situations with different one-step responses. The strict-realization test therefore used (r, c) as its primary comparability granularity.

The downstream E2 bits D_1, D_2 do not enter this local response. Consequently, adding or removing those bits does not change the (r, c) -based comparability relation, even though it can change equality of complete global configurations.

Strict dependency sites

For every $(r, c) \in \mathcal{I}$ and every frozen channel intervention a , the response was computed separately under the two receiver-memory contexts. A strictly realized dependency site was defined as

$$(a, r, c) \text{ such that } (r, c) \in \mathcal{I} \text{ and } X_0(a; r, c) \neq X_1(a; r, c).$$

The strict-site count was the number of such intervention/cell combinations.

For family 20, the procedure was applied independently to the two decisive edges. The corresponding strict counts and shared-cell counts are reported in Results Section 2.2. The method does not require the complete global state to coincide across memory contexts.

Realized-transition witnesses

A second analysis asked whether the contextual difference could be exhibited without applying a counterfactual overwrite.

For a local receiver state r and two channel values c_1, c_2 such that $(r, c_1), (r, c_2) \in \mathcal{I}$, choose realized orbit witnesses in each memory context and define the realized transition contrast

$$O_m(r; c_1, c_2) = e_B[T(s(r, c_1, m))] \oplus e_B[T(s(r, c_2, m))],$$

where e_B extracts the receiver module and $s(r, c, m)$ denotes any realized state with the indicated local coordinates. The contrast is well defined for this purpose because, conditional on m , the relevant one-step receiver response depends only on (r, c) ; globally different realized states sharing those local coordinates yield the same local contrast.

A realized-transition witness exists when

$$O_0(r; c_1, c_2) \neq O_1(r; c_1, c_2).$$

This witness is deliberately stricter in a different sense from full-fiber analysis: it uses only transitions occurring on the actual deterministic orbit, while matching the local state that is sufficient for the causal contrast.

Implementation audit

The strict-realization workstream used a read-only copy of the frozen instrument and the already-opened post-scoring confirmatory instance/key. The primary family-20 calculation and the broader battery were independently reimplemented and cross-checked against the frozen transition law. A previously interrupted run was archived and then re-executed; the family-20 output was byte-identical, and the complete re-execution agreed apart from runtime metadata. These checks are internal reproduction, not external scientific replication.

4.12 Information-loss refinement hierarchy

The post-confirmatory decomposition treated C1' as a sequence of representations of the same underlying response field:

$$\begin{aligned} \text{full XOR response pattern} &\rightarrow \text{pointwise Hamming weight} \rightarrow \text{row-resolved pairwise field} \\ &\rightarrow \text{within-pair row multiset} \rightarrow \widetilde{W} \rightarrow d \rightarrow \text{weak rank}. \end{aligned}$$

Not every arrow in this chain is lossy. In particular, for the frozen nine-intervention family, the map $\widetilde{W} \mapsto d$ is injective and is an isomorphism onto its admissible profile image (Section 4.9). The decomposition therefore localized where information disappeared **before** $\widetilde{W}$, and separately identified the ordinal loss at the final weak-rank step.

Four nested blind sets

For each decisive System-II edge, the two memory contexts were compared at four increasingly coarse resolutions.

Pattern-level blindness Define

$$B_{\text{pattern}} = \{\text{edges} : X_0(a; r, c) = X_1(a; r, c) \ \forall a, r, c\}.$$

These edges contain no memory-dependent difference even in the full pointwise XOR response pattern.

Row-resolved Hamming blindness Let

$$w_{m,r}(c_1, c_2) = \text{pc}_2(M[r][\pi_m(c_1)] \oplus M[r][\pi_m(c_2)]).$$

Define B_{row} as the set for which the row-resolved Hamming fields agree across contexts for all rows and channel pairs. Equivalently, every pointwise response has the same Hamming weight across contexts, although its bit pattern can differ.

Multiset blindness For each channel pair (c_1, c_2) , retain the multiset

$$\mathcal{H}_m(c_1, c_2) = \{\{w_{m,r}(c_1, c_2) : r = 0, \dots, 3\}\}.$$

Define B_{multiset} by equality of these multisets for every channel pair. This representation discards which receiver row produced which response weight but retains the within-pair distribution.

Aggregate-geometry blindness Finally,

$$\widetilde{W}_m(c_1, c_2) = \sum_r w_{m,r}(c_1, c_2).$$

Define

$$B_W = \{\text{edges} : \widetilde{W}_0 = \widetilde{W}_1\}.$$

By Section 4.9, B_W is exactly the L1 set for the frozen instrument.

These sets obey the refinement nesting

$$B_{\text{pattern}} \subseteq B_{\text{row}} \subseteq B_{\text{multiset}} \subseteq B_W.$$

In the studied 20,000 decisive System-II edges, all inclusions were strict and the observed cardinalities were

$$7 \subset 36 \subset 210 \subset 705.$$

The set inclusion is structural; the four numbers are generator-specific empirical counts.

Localization of L1 information loss

Of the 705 L1 edges, 7 already belonged to B_{pattern} and therefore had no pointwise memory dependence.

The remaining 698 L1 edges retained pointwise dependence. Their first loss of distinguishability occurred at three successive stages:

$$29 \quad (4.2\%)$$

became indistinguishable at the row-resolved Hamming stage,

$$174 \quad (24.9\%)$$

became indistinguishable only after row identity was discarded while retaining within-pair response multisets, and

$$495 \quad (70.9\%)$$

required arithmetic aggregation within the pairwise cells: the multisets differed but their sums agreed. Thus

$$705 = 7 + 29 + 174 + 495.$$

Family 20 lies in the final, dominant arithmetic-aggregation category on both decisive edges.

No cross-cell compensation in L1

Because the frozen d -profile is an injective linear representation of $\widetilde{W}$ (equivalently, the reduced six-coordinate map is invertible), equality $d_0 = d_1$ forces equality of every cell of $\widetilde{W}$. L1 cannot therefore be produced by a positive discrepancy in one pairwise cell cancelling a negative discrepancy in another.

A block-level audit confirmed this consequence in all 22,560 non-null blocks examined from the L1 edges: zero L1 cases required compensation between the pairwise/cell **sums** that determine $\widetilde{W}$. The cancellation responsible for the dominant L1 mechanism therefore resolves within a pairwise aggregation cell, below $\widetilde{W}$. This statement does not mean that every intermediate representation is blockwise invariant: among the 495 arithmetic-aggregation (B3) edges, 459 showed equality of some coarser intervention-level histograms through compensation between lower-level blocks, while the final pairwise sums still agreed cell by cell. Such intermediate histogram compensation is distinct from, and cannot create, L1 once $\widetilde{W}$ is fixed.

Ordinal loss is a separate stage

The transition

$$d_m \longrightarrow \rho_m$$

creates an additional blind set not contained in B_W . L2 edges have $d_0 \neq d_1$ but $\rho_0 = \rho_1$. In E1, the allowed cardinal movements of the affected duplicated pairing blocks occur in a discrete quantum of 64 under the frozen multiplicities; nevertheless, these changes can leave the weak order unchanged. The L2 mechanism is analyzed separately in Section 4.10.

4.13 Post-confirmatory generator-specific frequency study

This analysis was **not** part of the original confirmatory preregistration. It was initiated after the confirmatory failure to estimate how often the identified edge-level blind routes recurred under the same frozen generator law. Each generator-specific frequency batch was precommitted before its own generation and analysis.

Sampling law

The generator-specific frequency generator reused the frozen family law for the System-II core. The tables

$$F_0, G_0, H, K, \sigma_A, \sigma_B$$

were sampled uniformly over their respective finite codomains. The interface permutations π_0, π_1 were sampled independently and uniformly from S_4 , with rejection when

$$\pi_0 = \pi_1.$$

A sampled family was retained only if it satisfied the same core eligibility requirements used for the confirmatory nuclear family: E1-E5 for Systems I, II and III, together with E6 for System II.

Precommitted batches

Two batches were generated:

- seed 910000001: $N = 5000$ eligible System-II families, obtained after 17,135 attempts;
- seed 910000002: $N = 5000$ eligible System-II families, obtained after 16,845 attempts.

The combined population therefore contained

$$N_{\text{families}} = 10,000$$

and

$$N_{\text{edges}} = 20,000$$

decisive channel-to-processor edges.

For each edge, the two cardinal profiles d_0, d_1 , their canonical weak ranks, and pointwise dependency count were computed using the frozen one-step semantics.

The edge was assigned to exactly one level:

$$\text{L1} : d_0 = d_1,$$

$$\text{L2} : d_0 \neq d_1 \wedge \rho_0 = \rho_1,$$

$$\text{L3} : \rho_0 \neq \rho_1.$$

A System-II instance was counted as a **total C1' collapse** when both decisive channel-to-processor edges were STATE under C1', i.e. each was L1 or L2. This is the instance-level event that closes the four-edge STATE ring in the frozen System-II architecture.

Primary generator-specific frequency estimates

The combined edge counts were:

$$705/20,000 \text{ L1,}$$

$$225/20,000 \text{ L2,}$$

and

$$19,070/20,000 \text{ L3.}$$

Pointwise memory dependence was present in

$$19,993/20,000$$

edges.

At the instance level, the two batches produced 22 and 24 total collapses, respectively, for

$$46/10,000 = 0.0046.$$

The reported approximate 95% interval was the two-sided Wilson score interval,

$$0.003451 \leq p \leq 0.006130,$$

or approximately

$$0.345\%-0.613\%.$$

The 46 instances were partitioned into

$$29 \text{ L1/L1}, \quad 10 \text{ L1/L2}, \quad 7 \text{ L2/L2}.$$

These are frequencies under the frozen eligible generator law, not estimates of a universal or natural-system rate.

Relation between E1 and E2 scale

For the cardinal channel statistic, the two downstream E2 bits are inert with respect to the core and contribute only free multiplicity to the full fiber. Therefore

$$d_m^{E2} = 4 d_m^{E1},$$

and C1' weak ranks, L1/L2/L3 levels, and the resulting STATE/SIGNAL decisions transfer exactly between the $n = 10$ and $n = 12$ forms for the corresponding core family.

This scaling statement does **not** imply that every post-confirmatory quantity scales by four. In particular, complete-orbit structure and raw dependency-site counts involving additional free coordinates are distinct objects and were not generalized by this lemma.

Additional internal robustness samples

Separate precommitted OOS and coordinator checks used fresh seeds and independently implemented analysis routes. Their collapse estimates remained of the same order as the primary 0.46% estimate, including a fresh coordinator System-II sample of $N = 400$. These checks support recurrence within the frozen generator family but remain internal to the same research workflow and are not external scientific replication.

4.14 Temporal-composition analysis

This section describes an **Extended Results / Supplementary** post-confirmatory analysis. It does not alter C1', does not define C1'', and is not part of the confirmatory result.

The purpose was to test whether the exact one-step memory null found for System III is preserved when the same intervention/response construction is applied directly to

$$T^2 = T \circ T.$$

Horizon 1: exact System-III null

Consider the System-III edge $C_{AB} \rightarrow B$. In System III the receiver reads the interface directly, so for local state (y, m, c) ,

$$y_1 = G_0[y][c] \oplus \sigma_B[m],$$

$$m_{B,1} = K[m][y].$$

Under a channel intervention $c \rightarrow c^*$, the one-step XOR contrast in the processor bits is

$$G_0[y][c^*] \oplus G_0[y][c],$$

because the common additive term $\sigma_B[m]$ cancels. The current-step memory update does not depend on c , so its intervention contrast is zero.

Consequently, the complete one-step receiver response field is independent of m :

$$X_0^{(1)}(a; r, c) = X_1^{(1)}(a; r, c) \quad \forall a, r, c.$$

The same derivation holds symmetrically for $C_{BA} \rightarrow A$. Therefore, for eligible System III under the frozen one-step semantics,

$$\text{dep}^{(1)} \equiv 0, \quad d_0^{(1)} \equiv d_1^{(1)}.$$

This is the analytic result designated III-1. It is a property of the frozen one-step XOR-interventional field, not a statement that receiver memory is causally inactive in System III.

A post-confirmatory measurement generated 2,000 eligible System-III families with seed 910000020, giving 4,000 decisive edges. The analytic null was reproduced on all 4,000 edges.

Horizon 2: direct composition

For the same $C_{AB} \rightarrow B$ edge, write $s_B = \sigma_B$. After the first transition,

$$y_1 = G_0[y][c] \oplus s_B[m],$$

$$m_{B,1} = K[m][y],$$

and

$$c_{AB,1} = x.$$

After the second transition,

$$y_2 = G_0[y_1][x] \oplus s_B[m_{B,1}],$$

$$m_{B,2} = K[m_{B,1}][y_1].$$

Applying the original channel intervention $c \rightarrow c^*$ before the first transition and comparing after two transitions gives

$$\begin{aligned} X_{m,\text{proc}}^{(2)} &= G_0[G_0[y][c^*] \oplus s_B[m]][x] \\ &\quad \oplus G_0[G_0[y][c] \oplus s_B[m]][x], \end{aligned}$$

and the memory-bit contrast

$$\begin{aligned} X_{m,\text{mem}}^{(2)} &= K[K[m][y]][G_0[y][c^*] \oplus s_B[m]] \\ &\quad \oplus K[K[m][y]][G_0[y][c] \oplus s_B[m]]. \end{aligned}$$

The additive term applied **at the second transition** cancels between intervention and baseline because $m_{B,1} = K[m][y]$ is identical in the two branches. However, the first-step term $s_B[m]$ survives inside the row index of G_0 , and $K[m][y]$ selects the second-step memory-update row. The algebraic identity that removes m at horizon 1 therefore does not generally iterate under direct composition.

The symmetric derivation holds for $C_{BA} \rightarrow A$ with F_0, H, σ_A .

This is III-2a: the exact two-step expression no longer contains the algebraic cancellation identity that makes the one-step field memory-independent; receiver memory survives explicitly through future response-row indices and memory-update selectors. Thus the **one-step cancellation identity is not algebraically stable under direct composition to T^2** . This analytic statement does not assert that every eligible System-III edge must realize a nonzero context difference at horizon 2; that occurrence rate is an empirical question addressed below.

Empirical horizon-2 analysis

Before the horizon-2 executions, the workstream recorded an addendum specifying the direct T^2 construction and its checks. The analysis used the same 2,000 System-III families generated from seed 910000020; thus the horizon-2 observations are paired with, not statistically independent from, the horizon-1 sample.

For each of the 4,000 decisive edges, the full aligned-fiber two-step response was computed, and the same post-confirmatory dependency/L1/L2/L3 definitions were applied to the resulting T^2 field.

Memory dependence at horizon 2 was present in

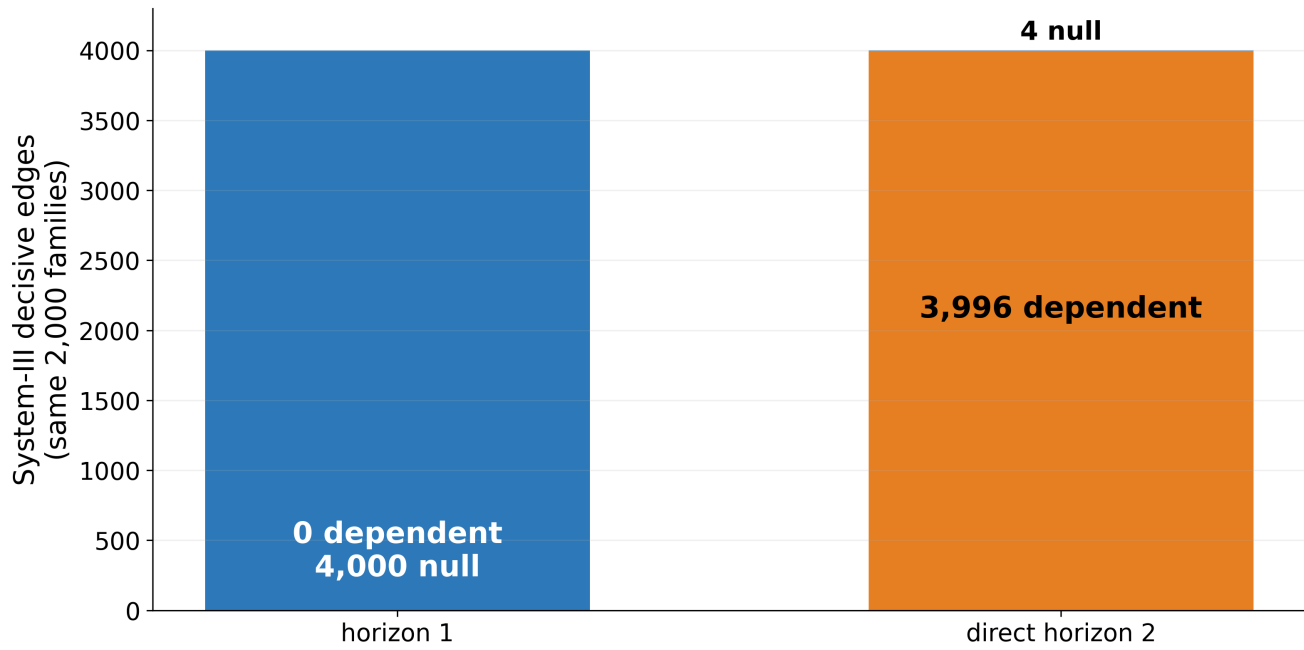
$$3996/4000 = 99.90\%$$

of the edges; four edges had zero dependency. All 2,000 instances had at least one memory-dependent decisive edge. The edge-level levels were

$$3334 \text{ L3}, \quad 632 \text{ L2}, \quad 34 \text{ L1}.$$

The exact closed-form two-step calculation used a multiplicity of $8 = 2^3$ after fixing the relevant outer processor state, receiver state, and channel value, and was checked against the transition machinery on all 4,000 edges.

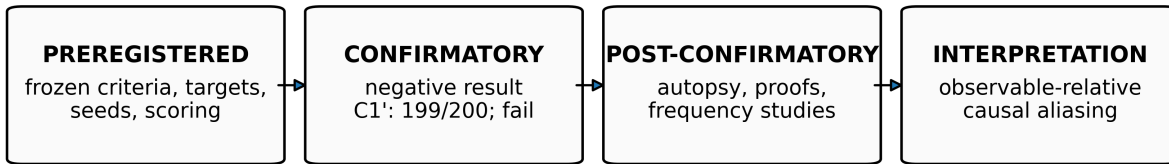
These frequencies constitute III-2b, an empirical post-confirmatory result under the studied generator law. They are not a theorem and do not imply that every horizon extension of C1' must fail. The supported conclusion is narrower: **directly applying the same class of response contrast to T^2 does not preserve the exact System-III null that holds at T** . A different functional using information jointly across horizons remains logically possible.



At $h = 2$: 3,334 L3 | 632 L2 | 34 L1. The 3,996/4,000 rate is empirical, not a theorem.

Figure S1. Temporal specificity of the System-III response field. The one-step response field is dependency-null on all 4,000 decisive edges in the paired 2,000-family sample. Under direct composition to T^2 , memory dependence is observed on 3,996/4,000 edges. The horizon-2 rate is empirical; the analytic result is the non-stability of the one-step cancellation identity under direct composition.

4.15 Internal verification, provenance, and reproducibility



Post-confirmatory analyses explain the failure; they do not alter or retroactively validate the confirmatory result.

Figure S2. Epistemic chronology. Post-confirmatory analyses explain the failure but do not alter or retroactively validate the frozen confirmatory result.

Frozen source record

The final v8.3 instrument consisted of 12 Python source files. Before the confirmatory execution, their SHA-256 digests were recorded in `hashes-finais-dev.txt`. The execution orchestrator independently recomputed the source hashes and aborted unless the complete source-file dictionary exactly matched that manifest.

The same 12-source manifest was later checked against the read-only copies used by the post-confirmatory workstreams. The complete `hashes-finais-dev.txt` manifest is the authoritative software identifier; individual

digests are not selectively reproduced here and should be supplied as a single machine-readable supplement.

Before confirmatory data generation, 50 tests across the five suites were green.

Execution environment capture

The preregistered execution orchestrator recorded the registered machine’s platform identifier, node identifier, full Python version string, and NumPy version to both the consolidated execution record and `hardware.txt` before generation. The final journal supplement should reproduce those values verbatim from that archived execution file rather than from a later environment.

The post-confirmatory multiagent analyses were run from `/root/prereg-env/bin/python`; the recorded workstream environment was Python 3.14.4 with NumPy 2.5.2. Post-confirmatory scripts were executed with bytecode writing disabled, and workstreams wrote only within their assigned analysis directories.

Confirmatory seeds and one-use rule

The closed execution record gives the two confirmatory root seeds as

$$S_{E1} = 3786434918, \quad S_{E2} = 3786434919.$$

That record states that each was consumed exactly once for its designated confirmatory generation. The present manuscript red-team did not regenerate either seed or reconstruct the hidden key from it. The procedural blinding contract prohibited regeneration of the hidden key from a confirmatory seed during classification.

The generator wrote the blinded instances and hidden keys separately. Before the classifiers were run, the keys were moved outside the classifier-accessible location. During the blinded analysis stage, inspection was restricted to operational metadata required to establish completion and integrity; the hidden scientific labels were not opened until the prespecified scoring step.

Blinding was therefore **procedural, not cryptographic**: possession of the frozen generator and root seed would in principle permit reconstruction. The claim made here is compliance with the predeclared workflow, not cryptographic impossibility of reconstruction.

Confirmatory execution checks

The confirmatory E1 and E2 generation steps completed successfully. Before scoring, the workflow verified:

- expected instance counts and unique IDs;
- exhaustive representational-equivalence enumeration;
- required E2 scale validity;
- the C3 implementation-canary condition;
- file and code integrity.

The closed confirmatory audit records an exhaustive equivalence check of

$$524,288/524,288$$

enumerated representations with zero discrepancies. The execution record also places the exhaustive equivalence stage within the preregistered 72-hour gate. Exact registered-machine wall-clock timings are archival provenance and should be copied verbatim from the original execution log in the reproducibility supplement rather than reconstructed from later analyses.

No scientific candidate result was inspected during the blind execution phase beyond the operational checks allowed by the protocol. The official confirmatory result was produced only after the key-opening/scoring stage.

Artifact immutability

The official confirmatory record was retained under

`/root/causal-A-amd2-official/`.

After closure, the core result artifacts were marked immutable at the filesystem level. The coordinator’s post-confirmatory audit verified the immutable attribute on `/root/causal-A-amd2-official/prereg-A/equiv-agregado.json`, `/root/causal-A-amd2-official/prereg-A/class-e1.json`, and `/root/causal-A-amd2-`

`official/resultado-pontuacao-A.txt`. The last file resides at the root of `causal-A-amd2-official/`, not inside `prereg-A/`; that path distinction was recorded as an audit-note correction and did not alter the artifact. The post-confirmatory workstreams imported read-only copies of the instrument and did not write into the official confirmatory directory.

Post-confirmatory precommitment and seed separation

Post-confirmatory analyses used seed ranges distinct from the confirmatory roots. Relevant precommitted or independently checked streams included

910000001-910000005, 910000010, 910000020, 910000030-910000033, and coordinator seeds 910000050-910000051.

The use of a post-confirmatory precommit fixes an analysis before **that post-confirmatory execution**; it does not convert the analysis into confirmatory evidence for the original preregistered hypothesis.

Internal reproduction taxonomy

Four different senses of independence were distinguished.

1. **Data-stream independence.** Some checks used fresh precommitted seeds rather than replaying the primary post-confirmatory population.
2. **Implementation independence.** Several workstreams reimplemented the relevant quantity using a route different from the frozen classifier or from another workstream—for example, the WS1 closed-form calculation, WS2 site-level decomposition, WS3 realized witnesses, WS5 audit implementation, and coordinator reconstructions.
3. **Internal analytic independence.** WS1 performed its derivation behind an analytic firewall before seeing the prior closed-form condition and independently arrived at an equivalent characterization.
4. **External scientific replication.** This has **not occurred**.

The second-pass WS4 and WS5 analyses had access to their first-pass materials and are therefore internal reanalysis/reproduction, not blind independent replication.

Workstream integrity checks

Each workstream verified its frozen-copy source hashes before analysis. Interrupted deterministic analyses in WS2 and WS3 were preserved and audited; re-execution produced byte-identical scientific outputs where identity was expected. The coordinator separately reconstructed the rank-6/determinant-8 system, rebuilt family 20 from the blinded representation, and used fresh System-II and System-III samples under new seeds. No violation of the frozen results or post-confirmatory theorems was found.

The final Phase-6 reports and their historical first-pass snapshots were themselves SHA-256 recorded. This permits the provenance chain

frozen instrument → confirmatory artifacts → post-confirmatory workstream outputs → Phase-6 synthesis

to be audited without rewriting earlier artifacts.

Archival packaging boundary

The scientific-content freeze of this Methods block does not substitute for packaging the original execution archive. Before journal submission, the reproducibility supplement should copy verbatim the registered-machine `hardware.txt` fields, the complete source-hash manifest, the original seed/one-use record, and the exact wall-clock timing log. Those archival strings were deliberately not reconstructed from the later post-confirmatory environment during this manuscript pass. If an archival record were ever found to disagree with the closed provenance summarized here, the discrepancy must be reported explicitly rather than silently reconciled.

Reproducibility scope

All claims of reproduction in this article refer to deterministic replay, independent implementation within the same research workflow, or fresh-seed OOS checks within the same frozen generator. None is presented as replication by an external laboratory, external analyst, or independently maintained codebase.

The confirmatory result remains the original negative result. The post-confirmatory analyses characterize the failure and its scope; they do not revise the preregistered decision.

5. Construct validity and scope

The target distinction is operational and fixed by construction

This study does not begin from the assumption that **STATE** and **SIGNAL** are universal natural kinds. It tests whether a frozen interventional observable preserves a **pre-specified mechanistic distinction in a controlled synthetic domain**.

Systems II and III were deliberately constructed to differ in how receiver state enters the incoming interface. In System III, the receiver reads the incoming channel directly. In System II, the retained receiver state selects a contextual permutation of that interface before the same processor mechanism is evaluated. The corresponding component partitions were fixed before confirmatory generation and scoring. **STATE**, **SIGNAL**, System II, and System III therefore name operational classes and targets of the generative design.

Accordingly, a target mismatch means that the classifier failed to preserve the distinction instantiated by that testbed under the prespecified scoring rules. It does **not** establish that an independently existing biological, physical, or computational ontology has been misclassified. Conversely, a target match would not by itself establish that the same labels provide a universal taxonomy outside the synthetic domain.

The confirmatory result is interpreted at exactly this level. C1' failed the preregistered conjunctive target because one E2 System-II instance produced the System-III-like operational partition; C2 and C3 also failed their scored targets. None of these outcomes is reinterpreted as a claim about universal STATE/SIGNAL ontology.

The counterexample is not only an architectural label mismatch

A stronger construct-validity concern is that the failed System-II target might encode a distinction present only in the generator's architecture, while the realized dynamics of the failed instance contain no causal difference for the classifier to recover. If that were true, the counterexample would mainly expose a mismatch between a human-assigned architectural label and the system's realized behavior.

The post-confirmatory autopsy rejected that explanation for the C1' counterexample. On both decisive channel-to-processor edges, the pointwise interventional response changes with receiver memory. Both memory contexts are reached by the deterministic trajectory. More strongly, when realized states are compared at the local granularity (r, c) that is response-sufficient for the frozen one-step contrast, both edges contain strictly realized context-dependent response differences. Realized-transition witnesses also exhibit the difference without requiring full-fiber counterfactual states.

This does not elevate the System-II label into a universal ontological truth. It establishes the narrower premise required for the paper's central interpretation: **there is an independently demonstrated causal difference below the observable that becomes invariant**. Without that premise, equality of the observable would constitute only blindness. With it, the family-20 failure is a demonstrated instance of observable-relative causal aliasing within the studied construction.

The result concerns an observable, not all causal information

The failed criterion observes the system through a particular sequence of representations. Pointwise response patterns are reduced to Hamming information, aggregated into a cell-weight geometry and cardinal intervention profile, and then coarsened again to a weak-order representation for the C1' edge decision.

The post-confirmatory refinement analysis shows that the resulting blind set is not intrinsic to all available causal information. Under progressively finer representations, the studied blind sets are strictly nested,

$$B_{\text{pattern}} \subset B_{\text{row}} \subset B_{\text{multiset}} \subset B_W,$$

with generator-specific cardinalities

$$7 \subset 36 \subset 210 \subset 705$$

among the 20,000 decisive System-II edges. Thus information that is invisible after cell aggregation can remain available at row-, multiset-, or point-pattern resolution.

For this reason, the paper does not claim global causal non-identifiability from all available causal information. The demonstrated limitation is **observable-relative**: a causal distinction can fail to be identifiable from a particular interventional representation even though finer causal information distinguishes the underlying response fields.

This distinction is essential to the interpretation. The result does not say that intervention is fundamentally unable to recover mechanism. It says that the evidential content of an intervention depends on what response information is retained by the observable applied to that intervention.

The analytic result has an explicit theorem class

The class-level symmetry result is broader than the particular family-20 instance but narrower than arbitrary causal dynamics. It applies when contextual response fields are pull-backs of a fixed response mechanism under interface relabelings, and the observable factors through the induced cell-weight geometry.

Within that class, if the relative interface permutation

$$\tau = \pi_1 \circ \pi_0^{-1}$$

belongs to the isometry group of the response pseudometric,

$$\tau \in \text{Iso}(W),$$

then any observable depending only on that geometry is blind to the contextual relabeling. This is a standard invariance consequence once the pull-back structure has been established.

The frozen System-II model supplies a separate model-to-theorem bridge: the one-step XOR contrast cancels the additive context term, the current-step memory update is independent of the intervened channel, and receiver memory enters the relevant response through π_m . Those facts establish that the frozen construction belongs to the theorem class.

The exact converse,

$$d_0 = d_1 \iff \widetilde{W}_0 = \widetilde{W}_1 \iff \tau \in \text{Iso}(W_M),$$

is narrower still. It relies on the full-rank measurement property of the frozen nine-intervention family and characterizes the metric L1 route only. It does not characterize the additional L2 route created when C1' coarsens cardinal profiles to weak order.

These nested domains prevent three distinct claims from being conflated:

1. symmetry invariance as general mathematics;
2. the contextual pull-back class in which that mathematics becomes causally relevant;
3. the exact L1 converse for the frozen intervention instrument.

Internal validity is stronger than external validity

Within the synthetic domain, several design features strengthen the **procedural and representational robustness** of the failure without independently validating the STATE/SIGNAL construct itself. The target partitions were fixed before confirmatory scoring; instances were exported under randomized bit and module labels; the confirmatory criterion was frozen; admissible scale projections and representational equivalences were tested; and the failure survived exhaustive representational-equivalence checks. The subsequent mechanistic analysis did not alter the official negative result.

The post-confirmatory mechanism was also checked through analytic reconstruction, deterministic replay, fresh-seed samples within the same generator family, and multiple internal implementations. These checks reduce the likelihood that the reported mechanism is a coding accident or an idiosyncrasy of one serialized representation.

They do **not** establish external validity. The empirical generator-specific frequency estimate of 46/10,000 total System-II collapses is a property of the frozen eligible generator law. The counts in the refinement hierarchy are likewise generator-specific. No biological system, engineered multi-agent system, physical network, or independently developed simulator has yet been shown to instantiate the same causal composition. No external scientific replication has occurred.

The strongest defensible scope is therefore asymmetric:

- the **analytic blindness proposition** extends to systems satisfying its explicit structural assumptions;
- the **exact L1 equivalence** extends only to instruments whose measurement operator separates the relevant relabeling orbit;
- the **strictly realized family-20 witness** establishes existence within the confirmatory construction;
- the **population frequencies** describe the frozen generator and are not transportable without new data.

Generalization and falsification require separate studies

Two different external-validation questions follow from the present result and should not be conflated.

A first study should test **generalization within the theorem class**. It would preserve the contextual pull-back structure while varying interface alphabet size, receiver-response tables, pairwise response kernels, orbit-separating intervention operators, and independent implementations. Such a study could preregister the prediction that relabelings inside the relevant response-geometry isometry group remain invisible to observables that factor through that geometry, and that suitable refinements recover distinctions when they break the symmetry responsible for blindness.

A second study should test the **boundary of the theorem class** by deliberately breaking its assumptions: allowing channel-dependent memory updates, removing XOR cancellation, replacing aligned full fibers, or introducing context effects that are not pure relabelings. These tests ask where the pull-back reduction and the symmetry-blindness result cease to apply.

The first is a generalization test; the second is a falsification/boundary test. Neither is part of the present confirmatory dataset, and neither should be retroactively used to strengthen the current evidence.

Implication for interpretation

The construct-validity conclusion is therefore deliberately narrower than a claim of universal causal individuation:

The study demonstrates that, in a controlled synthetic domain with a preregistered mechanistic distinction, measured interventional invariance can fail to preserve a causal difference on common realized support when the observable identifies response fields related by a symmetry of the representation it retains.

What remains open is how often this composition arises in other system classes, whether different experimental access can refine the observable enough to break the relevant blind set, and where the result ceases to hold when the structural assumptions of the theorem class are violated.

Those are external-validity questions. They motivate subsequent applications and Discussion; they are not results of the present study.

6. Relation to prior work

The present result sits at the intersection of several mature literatures. Its contribution should therefore not be framed as the discovery that symmetry can reduce observability, that causal abstraction can be lossy, that macro-level causal claims can depend on micro-realizations, or that distinguishability can depend on temporal horizon. Each of those principles has substantial prior art. The relevant question is narrower: whether prior work already contains the specific composition identified here, in which **receiver internal state generates a contextual relabeling of an interface, the relative relabeling aligns with a symmetry of the receiver mechanism’s own response pseudometric, and an interventional observable becomes invariant despite an independently demonstrated, causal difference on common realized support.**

Symmetry-induced unobservability is established prior art

The connection between symmetry and loss of observability is classical in control and dynamical-systems settings. Whalen et al. (2015) use group representation theory to study how explicit network symmetries affect nonlinear observability and controllability. Kolar, Rams, and Schöberl (2018) show that symmetry transformations can map distinct solutions to identical input-output trajectories, thereby proving non-observability. Mesbahi, Bu, and Mesbahi (2021) derive a Koopman-based relation between nonlinear-system symmetry and observability, including cases in which measurement functions preserve the same symmetries as the dynamics.

These works establish the generic ancestry of the implication

$$\text{symmetry} \longrightarrow \text{indistinguishability} / \text{loss of observability}.$$

Accordingly, the present article makes no novelty claim for invariance under a symmetry group, for quotienting by a symmetry, or for the fact that a measurement map can be blind to transformations it preserves.

The role of symmetry here is more specific. The relevant transformation is not posited as a global symmetry of the underlying dynamical system. It is generated contextually by the receiver’s retained state through the relative interface relabeling

$$\tau = \pi_1 \circ \pi_0^{-1},$$

and it is tested against a response pseudometric

$$W_M$$

induced by the concrete receiver mechanism. The symmetry becomes scientifically relevant only after the frozen System-II dynamics are shown to realize this contextual pull-back structure.

Lossy causal abstraction and aggregation are also established

A second body of work shows that causal information can be lost or rendered ambiguous by aggregation and abstraction. Zhu et al. (2024) demonstrate that different micro-realizations of the same macro-intervention can induce different macro-level causal effects, so macro causal relations may have to be interpreted with reference to micro states. Xia and Bareinboim (2025) explicitly address causal abstraction under lossy representations, including cases in which distinct low-level interventions with different downstream effects map to the same high-level intervention. Li, Kaba, and Ravanbakhsh (2025) characterize identifiability of causal models only up to abstractions determined by the available intervention family, while Madaleno, Misra, and Markham (2026) provide identifiability results for coarsened causal DAG models under interventional data.

This literature directly motivates a concession central to our interpretation:

aggregation can remove causally relevant distinctions, and interventional data need not identify a unique fine-grained causal representation.

The present study therefore does not claim to have discovered that coarse-graining or lossy representation can hide causal structure.

The difference lies in how the loss arises. In the mechanism studied here, the analyst does not begin by selecting a low-to-high abstraction that intentionally identifies two causal cases. Instead, receiver state changes how a finite interface is presented to a fixed response mechanism. The resulting contextual transformation becomes invisible only when its relative relabeling falls inside a symmetry of the response pseudometric retained by the observable. In the confirmatory counterexample, the response fields remain different below that aggregate, including on strictly realized local support.

Endogenous contexts can alter causal mechanisms

The idea that internal or endogenous system state can change causal relationships is likewise not new. Günther et al. (2024) explicitly study causal discovery with endogenous context variables, including internal states associated with context-specific changes in causal mechanisms. More broadly, causal representation-learning work has characterized ambiguities that remain under interventional data. For example, von Kügelgen et al. (2023) identify latent causal representations only up to ambiguities that cannot be resolved from the available interventional environments, and Li et al. (2025) formalize identification only up to an intervention-dependent abstraction.

These works occupy the generic statement

$$\text{internal/endogenous context} \longrightarrow \text{context-dependent causal structure}.$$

They do not, in the prior-art audit conducted for this project, supply the joint mechanism required here: receiver state selecting interface relabelings, a relative relabeling evaluated against a mechanism-induced response pseudometric, exact blindness of the retained interventional representation, and persistence of the causal difference below that representation.

Micro-realization dependence and the realized witness

The distinction between macro-level equivalence and micro-level causal difference is particularly close to the aggregation literature. Zhu et al. (2024) show that causal interpretation at an aggregate level can depend on which micro-state realizes a macro-intervention. That result is an important ancestor of the present paper’s construct-validity argument.

Our use of a realized witness is nevertheless narrower. The family-20 analysis does not infer causal aliasing merely from two different architectural descriptions mapping to the same aggregate. It separately establishes that the relevant response fields differ and that the difference is present on the deterministic trajectory at the response-sufficient local state (r, c) . The paper therefore uses the term **causal aliasing** only after establishing

$$X_0 \neq X_1$$

independently of the equality of the measured representation.

This is why the central result is not the generic statement that “macro causal structure depends on micro-realization.” It is a concrete mechanism by which a difference on common realized support becomes invisible to a particular interventional representation.

Finite-horizon distinguishability is not new

The temporal-composition result also has substantial ancestry. Hybrid and switching-system observability has long been formulated over finite horizons. Baglietto, Battistelli, and Scardovi (2007), for example, characterize controls that permit reconstruction of switching sequences in a finite-horizon setting. Related hybrid-systems work studies when internal variables or discrete modes become reconstructible only over sequences of observations.

Accordingly, the present article does not claim that temporal horizon generally affects observability or distinguishability.

The narrower System-III result is that the **same frozen XOR-interventional response field** has an exact algebraic memory null at one step, while the identity producing that null is not algebraically stable under direct composition to T^2 : memory re-enters through future row indices and memory-update selectors. The observed 3996/4000 horizon-2 dependence is empirical under the studied generator law, not a theorem. This remains a secondary technical result rather than the paper’s primary contribution and stays in Extended/Supplementary material under the frozen article hierarchy.

Restricted injectivity and orbit separation are established mathematical structures

The post-confirmatory measurement-operator formulation is not claimed as new mathematics. In linear inverse problems, uniqueness on a structured model is classically characterized by excluding nonzero admissible differences from the null space of the measurement map. A standard example is low-rank matrix recovery, where uniform uniqueness for rank- r matrices is controlled by whether the measurement null space contains a nonzero rank- $2r$ difference (Eldar, Needell, and Plan, 2012). Homomorphic sensing studies unique recovery under finite collections of linear transformations, including permutation-type transformations and coordinate projections (Tsakiris and Peng, 2019; Peng and Tsakiris, 2021), while orbit-recovery work asks when invariant features determine a group orbit (Bandeira et al., 2023).

Within that ancestry, the condition

$$\ker(\mathcal{A}_T) \cap \Delta\mathcal{O}(W_M) = \{0\}$$

is the restricted-injectivity / difference-set condition specialized to the finite relabeling orbit of the mechanism-induced response pseudometric. Its role in the present article is technical rather than a separate novelty claim: it isolates the property required of the intervention-summary operator from the causal mechanism that generates the orbit. The study-specific result is then that the frozen intervention family satisfies a stronger full-ambient-rank condition, and that this property persists for 18 of its 512 subsets.

The alignment theorem is instrument-specific

The L2 alignment result likewise builds on classical mathematics. The action

$$S_4 \rightarrow S_3$$

on the three perfect matchings, its kernel V_4 , and weak-order invariance are not new mathematical objects. The specific result concerns how that classical action interacts with the frozen intervention design: if the contextual transport fixes the unprobed perfect matching, an ordinal L2 event is impossible on that edge; L2 requires a misalignment capable of transporting the unprobed pairing value into a measured position while preserving weak rank.

The prior-art audit did not identify an antecedent with that complete instrument-specific implication. It is therefore treated as a **secondary technical novelty candidate**, not as new group theory.

Specific gap retained after the adversarial audit

The prior-art audit was explicitly adversarial: its purpose was to eliminate candidate novelty wherever an equivalent antecedent could be found. It identified strong ancestors for essentially every generic ingredient of the result:

symmetry	→ unobservability,
lossy representation	→ causal information loss,
aggregation	→ micro-realization dependence,
endogenous context	→ context-specific causal mechanism/relations,
interventional data	→ identification only up to equivalence,
longer horizon	→ different distinguishability.

What it did **not** identify was an essentially equivalent antecedent for the following joint composition:

receiver internal state → contextual interface relabeling → relative interface permutation τ
→ mechanism-induced response pseudometric $W_M \rightarrow \tau \in \text{Iso}(W_M)$
→ exact blindness of the retained interventional observable

while the underlying response fields remain causally different and that difference is separately demonstrated on realized support.

This composition is therefore the article’s **primary novelty candidate**. In the failed confirmatory instance, the blindness has an operational consequence: C1’ assigns the same STATE relation where the preregistered System-II target requires separation, even though the realized dynamics retain separating causal information.

The wording is deliberately limited. No equivalent antecedent was identified in the adversarial, multi-domain prior-art audit. That search does not establish that no such antecedent exists, and the manuscript does not claim “first discovery,” “first ever,” or proof of historical priority. Literature with remote terminology, unindexed work, theses, books, or an unrecognized mathematical isomorphism could still supply a closer antecedent. Peer review and independent specialist scrutiny remain additional tests of the novelty claim.

Positioning statement

The article should therefore be read as contributing a **specific causal composition**, not a new general theory of invariance:

Receiver-state-dependent causal modulation can become exactly invisible to an interventional observable when the contextual relabeling induced by receiver state lies in a symmetry of the response pseudometric generated by the receiver mechanism itself. In the confirmatory counterexample, the underlying causal difference remains strictly realized, so the measured invariance is an observable-relative alias rather than evidence of causal equality.

This positioning preserves the established ancestry of symmetry, abstraction, aggregation, endogenous context, and horizon-dependent observability while isolating the narrower mechanism revealed by the preregistered failure.

7. Limitations

The confirmatory test was negative, whereas the central mechanism was characterized post-confirmatorily

The most important epistemic limitation is chronological. The preregistered confirmatory experiment tested C1' and C2 as candidate operational discriminators and returned a negative result. The mechanism emphasized in this paper—receiver-state-dependent contextual relabeling becoming invisible through a symmetry of the receiver's response pseudometric—was identified only after scoring, during the autopsy of the single C1' confirmatory counterexample.

The subsequent algebraic derivations, strict-realization analysis, information-loss decomposition, L2 taxonomy, generator-specific frequency study, and temporal-composition analysis are therefore **post-confirmatory** with respect to the original hypothesis. They were motivated by the confirmatory failure, but not all were fitted to the single failed instance: later population and temporal executions used separately specified definitions, fresh or fixed seeds, and in several cases precommitments made before those particular analyses were run. Those safeguards strengthen the post-confirmatory evidence but do not retroactively convert the mechanism into a preregistered confirmatory discovery.

The confirmatory counterexample is sufficient to defeat the original conjunctive success criterion. It is not, by itself, a confirmatory test of the explanatory mechanism discovered from that failure. The strongest next validation would be a separate preregistered study that tests the P3 mechanism on newly generated system families or an independently implemented system class without using the original failure to define success after observing the new data.

The testbed is synthetic and its operational targets are fixed by construction

Systems II and III were deliberately designed to instantiate different interface mechanisms, and STATE/SIGNAL are classifier labels associated with prespecified component-partition targets. This gives the study strong control over the distinction being tested, but it limits construct generalization.

The paper does not establish that STATE and SIGNAL are universal natural kinds, that the System-II/System-III distinction exhausts possible interaction mechanisms, or that the target partitions correspond to an independently validated ontology in biology, physics, or artificial intelligence. A failure to preserve the synthetic target demonstrates a limitation of the tested observable relative to that controlled mechanistic distinction; it does not by itself establish how an arbitrary natural system should be individuated.

The post-confirmatory realized-dynamics analysis strengthens the counterexample by showing that the failed instance contains a causal difference below the aggregate rather than merely an architectural label difference. That evidence supports the paper's observable-relative interpretation inside the studied construction, but it does not remove the external construct-validity limitation.

The analytic theorem has a restricted structural domain

The class-level blindness proposition is not a theorem about arbitrary causal systems or arbitrary interventions. At the abstract level it assumes contextual pull-backs of a fixed response field and an observable that factors through the induced cell-weight geometry. The frozen System-II application adds model-specific conditions: aligned full-fiber comparisons, a finite interface whose context dependence is expressed as relabelings π_m , a fixed receiver response mechanism, a current-step memory update independent of the intervened channel, and XOR differencing that cancels the additive context term.

The exact L1 equivalence

$$d_0 = d_1 \iff \widetilde{W}_0 = \widetilde{W}_1 \iff \tau \in \text{Iso}(W_M)$$

is narrower still. The post-confirmatory measurement-operator result shows that the relevant design requirement is orbit-separation: $\ker(\mathcal{A}_{\mathcal{I}}) \cap \Delta\mathcal{O}(W_M) = \{0\}$. The frozen nine-intervention family satisfies the stronger condition of global injectivity on the six-dimensional four-symbol pairwise-array space, with rank 6 and determinant -8 ; four minimal six-intervention subsets retain that rank. These formal facts reduce dependence on the literal number nine, but they do not imply that arbitrary intervention families, alphabets, kernels, or receiver mechanisms satisfy the same separation condition. The measurement-operator formulation and the 512-subset rank audit were derived

after the original confirmatory score and after the earlier P3-focused prior-art audit; they are presented here as post-confirmatory linear-algebraic extensions, not as independent historical-priority claims.

Likewise, the concrete geometry W_M uses the frozen Hamming/popcount kernel. The class-level symmetry argument can be formulated for other pairwise kernels, but their blind groups and separating measurements must be derived separately. The present article therefore does not establish a universal causal metric or a universal isometry condition for causal aliasing.

The strict-realization result is local to the analyzed contrast

The claim that the hidden causal difference is strictly realized uses the local state (r, c) , not equality of the complete global configuration across receiver-memory contexts. This is justified analytically because, conditional on memory, the frozen one-step channel-to-processor response depends only on (r, c) . At that granularity, both decisive family-20 edges contain realized separating witnesses.

This result should not be generalized to arbitrary coarse-grainings or arbitrary causal questions. A different response functional, temporal horizon, or system architecture can require a different causally sufficient comparison state. The analysis also uses realized **support**, not visitation frequencies, so it establishes that separating local configurations occur, not how often they occur along a trajectory or how robustly they would be sampled under noise.

The realized-transition witnesses further show that the relevant context dependence is present on the deterministic orbit, but they do not imply that observational transitions alone generally identify causal boundaries.

The empirical frequencies are conditional on one eligible generator law

The generator-specific frequency analyses reuse the frozen System-II family law and eligibility filters. The estimate

$$46/10,000 = 0.46\%$$

and the associated L1/L2/L3 frequencies therefore describe the distribution **conditional on that synthetic generator and its eligibility criteria**.

The exact numerical rates can depend on how response tables, memory maps, contextual permutations, initial conditions, interface size, and eligibility constraints are sampled. The same limitation applies to the observed refinement counts

$$7 \subset 36 \subset 210 \subset 705$$

and to the decomposition of the 698 L1 edges with pointwise dependence into 29, 174, and 495 cases.

Fresh-seed and OOS checks within the workflow show that the phenomenon recurs under the same family law. They do not establish its frequency under other generator distributions, real biological systems, engineered multi-agent systems, or physical networks. The present study therefore supports **existence and within-generator recurrence**, not a domain-general generator-specific frequency estimate.

Only C1' received a mechanistic failure analysis

C2 made 74 confirmatory target errors and C3 made 75. Both also failed on family 20, but neither was subjected to the mechanistic post-confirmatory autopsy performed for C1'.

Accordingly, this paper should not imply that the response-geometry mechanism explains C2 or C3 failures, nor that the C1' taxonomy exhausts the failure modes of the other candidates. C3 additionally served an implementation-canary role and did not enter the scientific H_A decision.

The explanatory scope of the present article is therefore asymmetric: the confirmatory study reports all frozen candidate outcomes, while the detailed mechanism paper follows the C1' counterexample.

The blinding was procedural rather than cryptographic

The confirmatory workflow removed semantic system labels through state-bit and module randomization, separated the hidden key from classifier-accessible files, and required the key to remain unopened until the prespecified scoring stage. The classifier itself had no key-reading path.

However, this was procedural blinding, not cryptographic concealment. An actor with the frozen generator and confirmatory seed could in principle reconstruct hidden information. Module boundaries and retained-memory designations also remained visible because the classifier required them.

Thus the study demonstrates compliance with a prespecified blind-analysis workflow, not information-theoretic impossibility of reconstructing the generating labels. One possible strengthening in a future replication would be to use independently generated hidden instances or a third-party key custodian so that the analysis team cannot regenerate the hidden mapping from the generative source.

Internal reproduction is not external replication

The post-confirmatory mechanism was subjected to multiple checks: deterministic re-execution, fresh-seed samples, alternative implementations, coordinator reconstructions, and an analytic firewall for the WS1 derivation. These checks materially reduce the probability that the result is a coding artifact inside the documented workflow.

They nevertheless share the same overall research program, synthetic family, and substantial portions of the same conceptual setup. Later WS4/WS5 passes also had access to earlier work. No external group has reproduced the theorem-to-model mapping, implemented the analysis independently from the manuscript, or tested the mechanism in a separately designed system family.

External replication therefore remains a major evidence gap. In particular, the primary novelty candidate would be substantially stronger if a new preregistered study predicted symmetry-induced blindness and symmetry-breaking recovery in a system family not derived directly from the original generator.

Observable refinement is not guaranteed to solve identification

The nested blind sets show that, in this studied class, retaining finer response information can strictly recover distinctions lost by coarser observables. This establishes that the demonstrated blind class is observable-relative rather than an intrinsic absence of all causal information.

It does **not** establish that arbitrary ambiguities can always be removed by adding measurements, nor that there exists a finite practical refinement sufficient for every system. Real systems may contain latent variables, inaccessible internal states, stochasticity, measurement noise, or equivalence classes that persist even under substantially richer instrumentation.

The appropriate conclusion is therefore conditional: observable refinement is a diagnostic route that can test whether a reported invariance is representation-induced. It is not a general guarantee of causal identifiability.

Temporal conclusions are narrow

The System-III horizon result is secondary. At one step, the frozen XOR-interventional field has an exact algebraic memory null; under direct composition to T^2 , the identity responsible for that null is not algebraically stable, and memory dependence was observed in 3996/4000 studied edges.

Several limitations follow. The horizon-2 sample used the same 2,000 underlying System-III families as the horizon-1 analysis, so it is a paired extension rather than an independent population. Four horizon-2 edges remained dependency-null. Only $h = 1$ and direct $h = 2$ composition were characterized. The result therefore does not support a universal claim that longer horizons improve individuation, that every System III becomes distinguishable at $h = 2$, or that no multi-horizon functional can preserve the desired distinction.

Novelty remains a candidate, not a proved historical priority

The adversarial prior-art audit found strong antecedents for all generic ingredients of the result-symmetry-induced unobservability, lossy causal abstraction, micro-realization dependence, endogenous causal contexts, equivalence-class identifiability, and finite-horizon distinguishability. It did not identify a T0 antecedent for the specific P3 composition.

That search is evidence for positioning, not a proof that no equivalent antecedent exists. Bibliographic indexing is incomplete; terminology can differ radically across fields; theses, books, older literature, or an unnoticed mathematical isomorphism may contain a closer result. The prior-art audit was also conducted within the same research program and does not replace independent expert review.

For this reason the article uses **primary novelty candidate** rather than “first discovery” or “proved novel.” A plausible T0 antecedent identified during peer review should reopen the positioning audit rather than be treated as a cosmetic citation update.

Real-world relevance remains prospective

The biological and artificial-system scenarios discussed in this article are motivations for external validation, not demonstrations. No biological network, AI system, multi-agent architecture, or physical device was tested.

Likewise, the speculative possibility that intervention itself could induce compensatory equivalence through feedback or adaptation is distinct from the representation-induced aliasing characterized here. It has not been tested, formally derived, or subjected to the paper’s prior-art audit.

The present evidence therefore supports a precise methodological warning inside a defined theorem class and synthetic testbed. Whether that warning becomes frequent or consequential in particular real-world domains remains an empirical question.

Author contributions and data availability

Author contributions. Ruben Brito conceived the research programme and the causal-individuation problem, developed the theoretical framework, authored the preregistered protocol and amendments, defined the operational targets and evaluation criteria, directed the confirmatory and post-confirmatory analyses, interpreted the results, and wrote and revised the manuscript. The author takes full responsibility for the scientific content and conclusions.

Computational support. Computational tools supported software implementation, numerical verification, consistency checks, reference organization, and editorial refinement. Study-design decisions, scientific interpretations, claim selection, and final manuscript decisions were made by the author.

Data and code availability. The frozen twelve-file confirmatory instrument, preregistration and amendments, source-code hash manifest, confirmatory artifacts, post-confirmatory analysis records, seeds, and reproducibility documentation are retained in the project archive. The complete reproducibility package will be deposited in a permanent public repository or registration record upon submission, with the corresponding persistent identifier reported in the final published version.

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Supplementary Information

Supplementary Note A. Formal status and notation

The supplementary material preserves the frozen identifiers because they provide a direct audit trail to the preregistration and executable instrument. The main manuscript uses descriptive names first. Unless explicitly stated otherwise, all analytic results in this supplement are post-confirmatory and do not change the confirmatory score.

For any observable F , define the observable-induced equivalence relation

$$X \sim_F X' \iff F(X) = F(X').$$

An equivalence class under $\sim_F$ is the most general representation of observable indistinguishability. A group description is stronger and requires a group action. In the exact-profile route studied here, the relevant action is the permutation action of the finite interface alphabet and the symmetry group is $\text{Iso}(W_M)$.

Lemma A1. W_M is a pseudometric

For each receiver row r , the map $p \mapsto M[r][p]$ takes interface symbols into a finite Hamming space. Therefore

$$W_{M,r}(p, q) = \text{pc}_2(M[r][p] \oplus M[r][q])$$

is the pull-back of the Hamming metric and is a pseudometric on the interface alphabet. The sum $W_M = \sum_r W_{M,r}$ is also a pseudometric. Distinct symbols may have zero distance if every receiver row maps them to the same output, so *pseudometric* is the generic term; in non-degenerate instances it is a metric.

Supplementary Note B. Local-response reduction and realized support**Proposition B1. Response-sufficient local-state reduction**

Consider the System-II edge $C_{AB} \rightarrow B$. For a fiber state with receiver processor value r , channel value c , receiver memory m , and a transient intervention replacing c by c' , the frozen update gives

$$y' = G_0[r][\pi_m(c)] \oplus \sigma_B[m], \quad m'_B = K[m][r].$$

The XOR contrast between intervened and null transitions is

$$X_m(r, c; c') = (G_0[r][\pi_m(c')] \oplus G_0[r][\pi_m(c)], 0).$$

The additive term cancels, and the memory-update coordinate is unchanged because $K[m][r]$ is independent of c . Hence, conditional on m and the intervention, the response depends on the global state only through (r, c) . The same derivation holds for $C_{BA} \rightarrow A$ with F_0 in place of G_0 .

This proposition justifies (r, c) as the response-sufficient comparison state for the analyzed one-step contrast. It does not assert that (r, c) is a universal sufficient statistic for arbitrary functionals or horizons.

Common realized support

Let

$$\mathcal{C}_m = \{(r, c) : (r, c, m) \text{ occurs on the deterministic orbit}\}, \quad \mathcal{I} = \mathcal{C}_0 \cap \mathcal{C}_1.$$

A strict dependency site is an intervention-cell triple (a, r, c) with $(r, c) \in \mathcal{I}$ and $X_0(a; r, c) \neq X_1(a; r, c)$. Family 20 contains nine such sites on each decisive edge. The shared realized cells are

- $C_{AB} \rightarrow B$: $(0, 1), (2, 0), (3, 1), (3, 3)$;
- $C_{BA} \rightarrow A$: $(0, 0), (0, 3), (1, 0), (1, 3), (3, 2)$.

The full global-state intersection is empty in the E2 embedding, illustrating why response-irrelevant downstream bits must not define comparability for this one-step local response.

Supplementary Note C. Proof of Proposition 1 (symmetry-induced blindness)

Let $\widetilde{W}_m(c_1, c_2) = W_M(\pi_m(c_1), \pi_m(c_2))$ and $\tau = \pi_1 \circ \pi_0^{-1}$. If $\tau \in \text{Iso}(W_M)$, then for every pair (c_1, c_2) ,

$$\begin{aligned}\widetilde{W}_1(c_1, c_2) &= W_M(\pi_1(c_1), \pi_1(c_2)) \\ &= W_M(\tau\pi_0(c_1), \tau\pi_0(c_2)) \\ &= W_M(\pi_0(c_1), \pi_0(c_2)) \\ &= \widetilde{W}_0(c_1, c_2).\end{aligned}$$

Therefore every observable F that factors through $\widetilde{W}_m$ has equal values in the two contexts. The proof is purely an invariance argument; it establishes no underlying causal difference. That difference is an additional empirical/analytic premise supplied by the family-20 realized-support analysis.

Supplementary Note D. Measurement-operator criterion, exact-profile theorem, and frozen-design rank audit

Let $\mathcal{V}_C$ be the vector space of symmetric zero-diagonal pairwise arrays on interface alphabet C , and let $\mathcal{A}_I : \mathcal{V}_C \rightarrow \mathbb{R}^k$ be a linear measurement operator. If $d_m = \mathcal{A}_I(\widetilde{W}_m)$, linearity gives

$$d_1 - d_0 = \mathcal{A}_I(\widetilde{W}_1 - \widetilde{W}_0).$$

Therefore

$$d_0 = d_1 \iff \widetilde{W}_1 - \widetilde{W}_0 \in \ker(\mathcal{A}_I).$$

Both contextual geometries belong to the relabeling orbit $\mathcal{O}(W_M)$. For any $U, V \in \mathcal{O}(W_M)$, $\mathcal{A}_I(U) = \mathcal{A}_I(V)$ if and only if $U - V \in \ker(\mathcal{A}_I)$. Therefore the restriction of the measurement operator to the orbit is injective if and only if its kernel intersects the orbit-difference set only at zero. Under that condition,

$$\ker(\mathcal{A}_I) \cap \Delta\mathcal{O}(W_M) = \{0\},$$

measurement equality forces $\widetilde{W}_0 = \widetilde{W}_1$. By the contextual-pullback identity, the latter is equivalent to $\tau \in \text{Iso}(W_M)$. This proves Proposition 2. The condition is mechanism-relative: full injectivity of $\mathcal{A}_I$ on all of $\mathcal{V}_C$ is sufficient but stronger than necessary.

For $|C| = q$, $\dim \mathcal{V}_C = q(q-1)/2$, proving Corollary 1. In the frozen four-symbol design this dimension is six.

Write the six independent off-diagonal entries of the two-bit interface pseudometric as

$$\mathbf{w} = (w_{01}, w_{02}, w_{03}, w_{12}, w_{13}, w_{23})^\top.$$

The frozen intervention family measures

$$A = w_{01} + w_{23}, \quad B = w_{02} + w_{13},$$

and the four row sums

$$\begin{aligned}V_0 &= w_{01} + w_{02} + w_{03}, \\ V_1 &= w_{01} + w_{12} + w_{13}, \\ V_2 &= w_{02} + w_{12} + w_{23}, \\ V_3 &= w_{03} + w_{13} + w_{23}.\end{aligned}$$

Thus $\mathbf{s} = (A, B, V_0, V_1, V_2, V_3)^\top = L\mathbf{w}$ with

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}.$$

Direct integer elimination gives

$$\text{rank}(L) = 6, \quad \det(L) = -8.$$

Hence the measurement map is injective for the six independent entries of $\widetilde{W}$. For the frozen statistic,

$$d_m(a) = 2^{n-5} \sum_c \widetilde{W}_m(c, \text{sub}_a(c)),$$

so equality $d_0 = d_1$ is equivalent to $\widetilde{W}_0 = \widetilde{W}_1$. Because $\widetilde{W}_1$ is the pull-back of W_M by π_1 and $\widetilde{W}_0$ by π_0 , equality of the two is equivalent to $\tau \in \text{Iso}(W_M)$. This proves Theorem 1.

For completeness, the frozen nine intervention rows were also enumerated as an exact design audit. The null row has rank zero; the two low-bit assignments are duplicates of the A row; the two high-bit assignments are duplicates of the B row; and the four full-channel assignments supply the four V_i rows. Exact integer rank over all $2^9 = 512$ subsets yields 4 rank-6 subsets of size 6, 8 of size 7, 5 of size 8, and 1 of size 9. Thus 18 subsets are full rank in total. The four minimal subsets are precisely

$$\{c := 0, c := 1, c := 2, c := 3\} \cup \{b_0 := \alpha\} \cup \{b_1 := \beta\}, \quad \alpha, \beta \in \{0, 1\}.$$

This enumeration is a post-confirmatory algebraic property of the prespecified intervention family. It uses no confirmatory outcomes and does not alter the scoring rule.

The numerical form $d_m = 2^{n-5}(0, A, A, B, B, V_0, V_1, V_2, V_3)$ also yields the exact embedding relation $d_{E2} = 4d_{E1}$ for the same core family. This scaling preserves exact profiles and weak ranks; it does not imply that arbitrary orbit sizes or raw dependency-site counts scale by four.

Supplementary Note E. Ordinal blindness and the alignment theorem

The four interface symbols define three perfect matchings

$$\lambda_1 = \{01|23\}, \quad \lambda_2 = \{02|13\}, \quad \lambda_3 = \{03|12\}.$$

The action of S_4 on these matchings induces a homomorphism $\psi : S_4 \rightarrow S_3$ with kernel V_4 . Two matching sums are directly probed by the bit interventions; the third, denoted λ , is unprobed at that stage and is recoverable through the row sums in the exact-profile theorem.

If $\psi(\tau)$ fixes λ , the exclusion of L2 follows algebraically. Let μ and ν denote the two probed perfect matchings and λ the unprobed matching. For a pairwise array W , write $S_W(\eta)$ for the sum of the two edge weights in matching η , and $D_W(i) = \sum_{j \neq i} W(i, j)$ for the degree of interface symbol i . Up to the common multiplicity, the frozen cardinal profile is

$$q(W) = (0, S_W(\mu), S_W(\mu), S_W(\nu), S_W(\nu), D_W(0), D_W(1), D_W(2), D_W(3)).$$

For the contextual pull-back τ^*W , matching sums and degrees obey

$$S_{\tau^*W}(\eta) = S_W(\tau\eta), \quad D_{\tau^*W}(i) = D_W(\tau i).$$

If $\psi(\tau)$ fixes λ , then τ permutes the set $\{\mu, \nu\}$. Consequently $q(\tau^*W)$ is a coordinate permutation of $q(W)$: the two duplicated matching blocks are either fixed or exchanged and the four degree coordinates are permuted. The two profiles therefore have exactly the same multiset of cardinal values.

Now let $\rho(x)$ denote the canonical weak-rank vector of a finite vector x . If two vectors x and y have the same multiset of values and $\rho(x) = \rho(y)$ coordinatewise, then $x = y$. Indeed, write their common distinct values as $v_0 < \dots < v_t$. Canonical rank k identifies the numerical value v_k in both vectors, so equal rank at every coordinate forces equal cardinal value at every coordinate. Hence, in the aligned region,

$$\rho(d_0) = \rho(d_1) \implies d_0 = d_1.$$

Thus L2 is impossible whenever $\psi(\tau)$ fixes λ ; any L2 event requires misalignment. This implication is analytic and does not depend on the empirical census.

The census provides a separate verification of the predicted blind-zone signature. In the primary 20,000-edge population, all 225 L2 edges followed it: 193 T_out/FC_oth edges changed exactly one duplicated probed-pairing block, whereas 32 three-cycle (Cyc3) cell edges changed both in the predicted double exchange; the full-channel intervention block was unchanged in all 225. Across 21,390 aligned edges accumulated in the primary and validation analyses, zero L2 events were observed.

Supplementary Note F. Information-loss hierarchy

The frozen response-order decision can be represented as the chain

$$\begin{aligned} \text{full XOR response pattern} &\rightarrow \text{pointwise Hamming weight} \rightarrow \text{row-resolved pairwise field} \\ &\rightarrow \text{within-pair row multiset} \rightarrow \widetilde{W} \rightarrow d \rightarrow \text{canonical weak rank.} \end{aligned}$$

Define four equality-based indistinguishability sets: B_{pattern} , B_{row} , B_{multiset} , and B_W , where B_W is the L1 set. In the studied 20,000 decisive System-II edges,

$$B_{\text{pattern}} \subset B_{\text{row}} \subset B_{\text{multiset}} \subset B_W$$

with observed cardinalities

$$7 \subset 36 \subset 210 \subset 705.$$

Among the 698 L1 edges with nonzero pointwise memory dependence, 29 lost distinguishability at pointwise Hamming weight, 174 after loss of response location while retaining within-cell multisets, and 495 only under within-cell arithmetic aggregation. Family 20 belongs to this last, dominant category on both decisive edges. Equality of $\widetilde{W}$ is cellwise; the injective map $\widetilde{W} \leftrightarrow d$ excludes compensation between final measured cell sums as a source of L1.

Supplementary Note G. Generator-specific frequency study

Two post-confirmatory batches were precommitted before their own generation and analysis:

Seed	Eligible System-II families	Attempts
910000001	5,000	17,135
910000002	5,000	16,845

Across 20,000 decisive edges, 705 were L1, 225 L2, and 19,070 L3; pointwise memory dependence was present in 19,993 edges. At the instance level, 46/10,000 families collapsed completely under the response-order criterion: 29 L1/L1, 10 L1/L2, and 7 L2/L2. The Wilson 95% interval for the instance-level frequency is approximately 0.345%-0.613%.

These quantities are conditional on the frozen eligible generator distribution. They are not estimates for biological, artificial, or physical systems and are not universal constants of the theorem class.

Supplementary Note H. Temporal composition in the direct-interface benchmark

At horizon one in System III, the channel enters the receiver directly and the additive memory term cancels from the frozen XOR contrast. The current-step memory update is channel-independent. Therefore the one-step field is exactly memory-independent for the analyzed channel-to-processor contrast.

Under direct composition to T^2 , the first-step additive memory term survives inside the second-step response-row index and the first-step memory update selects the second-step update row. For $C_{AB} \rightarrow B$, the processor part of the two-step intervention contrast has the form

$$G_0[G_0[y][c^*] \oplus s_B[m]][x] \oplus G_0[G_0[y][c] \oplus s_B[m]][x],$$

and the memory component contains

$$K[K[m][y]][G_0[y][c^*] \oplus s_B[m]] \oplus K[K[m][y]][G_0[y][c] \oplus s_B[m]].$$

Thus the algebraic identity responsible for the one-step null is not stable under direct two-step composition. This does not prove nonzero dependence for every concrete edge. Empirically, on 2,000 paired System-III families (4,000 decisive edges; seed 910000020), 3,996 edges showed horizon-two memory dependence: 3,334 L3, 632 L2, and 34 L1; four remained dependency-null. No universal claim about arbitrary longer horizons or multi-horizon functionals is made.

Supplementary Note I. Frozen source manifest and execution provenance

The confirmatory instrument consists of twelve Python source/test files. The recorded development-freeze SHA-256 manifest is reproduced below; direct recomputation from the frozen archive matches every digest.

File	SHA-256
benchmark.py	19d0baf0e3a8742d44039893c34b8941da936144de1efdd9b21da27baa11f6a
classificador.py	ecaa40c6fa2abf84751811cbe5490073bb68e790ab1d5e135ea9342256b046a3
dryrun.py	43473e07e65d268a285208a410edc14ae69dc93a7a6e572ca74a3957dd54fd21
equivalencias.py	c87cdb38da3b468488671dc7f752cd9d7e06effaf9485ece937e9fa2716ec389
escala.py	40ccd532e583b05665b2fbed8bd2a63f18a4300615f6dba7fb3eaaa73e05cca7
gerador.py	6389c5615fd88540676164394d3bab0c572e85fffaa1a1ba3a300a13ddd58ccd
pontuacao.py	028f5a0327c3e3437ffc6df03d10d45d9a06c3d0ff267795caa6f9e751fe0f57
test_classificador.py	fb42d9f14bd50f3f40cd293e4f842e87844bedd79218036f8826ba18312e1528
test_equivalencias.py	461521efe3b3daf99a3d5a7a3851acedaf55be28ff58af48fb5c0b6128a475ce
test_escala.py	98a7543423757f0610c1bedd66a35638f48437baeea60df8ac03836164433f17
test_gerador.py	ebfda4d8967cf9e3cecf087a1fcacf901a2e5c735f2e099112f5f39ade9cc4254
test_pontuacao.py	905db9b5077aa8264554311284c5bba425a474848b2b780152e5374650ebf055

Fifty tests across five suites were green before confirmatory generation. The confirmatory root seeds were $S_{E1} = 3786434918$ and $S_{E2} = 3786434919$, each recorded as one-use for the designated generation. The confirmatory audit records exhaustive representational-equivalence enumeration of 524,288/524,288 representations with zero discrepancies and completion within the preregistered 72-hour gate. The original orchestrator recorded the registered machine’s platform, node identifier, Python version, NumPy version, and wall-clock timing to the execution archive; those strings should be reproduced verbatim from the original `hardware.txt` and timing log in a public submission archive rather than reconstructed from the later post-confirmatory environment. Post-confirmatory workstreams recorded Python 3.14.4 and NumPy 2.5.2.

Supplementary Note J. Evidence-status ledger for the main claims

Claim	Evidence status	Scope
confirmatory outcome is negative family 20 contains memory-dependent response differences on common realized (r, c) support	confirmatory immutable fact post-confirmatory demonstrated + internal reproduction	frozen Battery A confirmatory counterexample
$\tau \in \text{Iso}(W_M)$ implies blindness for the cell-factorizing class measurement-operator orbit-separation criterion $\ker(\mathcal{A}_T) \cap \Delta\mathcal{O}(W_M) = \{0\}$	analytic proposition post-confirmatory analytic proposition	stated contextual-pullback class linear intervention summaries on the relabeling orbit of the mechanism-induced pairwise geometry
frozen intervention subset audit: 18 rank-6 subsets, including 4 minimal six-intervention subsets	exact post-confirmatory algebraic design audit	frozen nine intervention measurement rows; no confirmatory outcomes used

Claim	Evidence status	Scope
$d_0 = d_1 \iff \widetilde{W}_0 = \widetilde{W}_1 \iff \tau \in \text{Iso}(W_M)$	analytic theorem + finite verification	frozen nine-intervention family, one-step channel-to-processor statistic
alignment theorem excludes L2 when $\psi(\tau)$ fixes the unprobed matching	analytic theorem + census checks	frozen 2-bit intervention geometry
46/10,000 total collapses	post-confirmatory generator-specific frequency	frozen eligible System-II generator
refinement counts 7/36/210/705	post-confirmatory census with structural nesting	studied 20,000 edges
3,996/4,000 horizon-two dependence	post-confirmatory empirical result	paired System-III sample, direct T^2
no essentially equivalent P3	closed adversarial prior-art audit	search current through August 2026;
antecedent identified		not proof of historical priority
biology/AI relevance	prospective interpretation only	untested externally